Math 210B Notes

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Chapter 1

Introduction to Rings

1.1 Ring and Field Definitions

We move on from studying groups to studying rings and fields. First, lets compare some analogues between groups and rings.

Groups:

(i) 1 operation
(ii) Subgroups
(iii) Normal groups N
(iv) Quotient groups G/N
(v) Morphisms of groups

Rings:

(i) 2 operations
(ii) Subrings
(iii) Ideals I
(iv) Quotient rings R/I
(v) Morphisms of rings

We build the theory of rings and fields in a similar way to the theory of groups. An important type of ring we wish to study is the ring of polynomials with coefficients in a field. Our goal is to be able to study Galois Theory and make a connection between automorphisms of fields and their subfields.

Before we get started, consider the following example.

Example 1.1.1. Let R be a set with operations $+$ and $\times$ such that distribution holds for all elements in the set:

$$\begin{aligned} \forall a, b, c \in R \quad a \times (b + c) &= (a \times b) + (a \times c) \\ (a + b) \times c &= (a \times c) + (b \times c) \end{aligned}$$

Further assume $+$ and $\times$ are associative and that there exists $1 \in R$ such that $1 \times a = a \times 1 = a$ for all $a \in R$. Let $-a$ and $-b$ be the additive inverses of $a, b \in R$. Show that

$$a + b = b + a$$

Proof.

$$\begin{aligned} (a + b) \times (1 + 1) &= (a + b) \times 1 + (a + b) \times 1 \\ &= a \times 1 + b \times 1 + a \times 1 + b \times 1 \\ &= a + b + a + b \end{aligned} \tag{I}$$

$$\begin{aligned} (a + b) \times (1 + 1) &= a \times (1 + 1) + b \times (1 + 1) \\ &= a \times 1 + a \times 1 + b \times 1 + b \times 1 \\ &= a + a + b + b \end{aligned} \tag{II}$$

From (I) and (II), we have

$$\begin{aligned} a+b+a+b &= a+a+b+b \\ \implies a+b+a+b-b &= a+a+b+b-b \\ \implies -a+a+b+a+0 &= -a+a+a+b+0 \\ \implies 0+b+a &= 0+a+b \\ \implies b+a &= a+b \end{aligned}$$

□

This example motivates the following definition.

Definition 1.1.1. (Ring)
A ring is a set R together with two binary operations $+$ (called addition) and $\times$ (called multiplication) satisfying the following:

(i) $(R,+)$ is an abelian group

(ii) Multiplication is associative $\forall a,b,c \in R$

$$(a \times b) \times c = a \times (b \times c)$$

(iii) Distributive laws hold $\forall a,b,c \in R$

$$\begin{aligned} \text{Left distribution: } & a \times (b \times c) = (a \times b) + (a \times c) \\ \text{Right distribution: } & (a+b) \times c = (a \times c) + (b \times c) \end{aligned}$$

If multiplication is commutative, we call R a commutative ring. The ring R is said to have an identity denoted 1 (or contains a unity element) if

$$1 \times a = a \times 1 = a \quad \forall a \in R$$

In this case, R is called a ring with unity.

Notation . $a \times b$ will be written as ab. The additive identity of $(R,+)$ will be denoted 0. The additive inverse of an element $a \in R$ will be denoted $-a$.

Notice that our definition for a ring does not require the existence of a multiplicative inverse for each element in the ring. The addition of multiplicative inverses leads to more specific types of rings, and with the addition of multiplicative commutativity, we get fields.

Definition 1.1.2. (Division Ring, Field)
A ring R with unity 1 (where $1 \neq 0$) is called a division ring if every $a \in R$ where $a \neq 0$ has an element $b \in R$ such that $ab = ba = 1$. That is, if all nonzero elements have a multiplicative inverse. If R is also commutative, then R is called a field.

Example 1.1.2. (i) Trivial rings: Given any group $(G,*)$ if we take $*$ as addition and define multiplication as $ab = 0 \ \ \forall a,b \in G$, then this forms a ring.

(ii) If $R = \{0\}$, this is called the zero ring with multiplication and addition defined as $0 \cdot 0 = 0$ and $0+0=0$. Note that this is the only ring where $1 = 0$. Show that if $1 = 0$, then $R = \{0\}$.

Proof. Let $a \in R$.

$$\begin{aligned} a \cdot 0 = a(0+0) &= a \cdot 0 + a \cdot 0 \\ \implies a \cdot 0 &= a \cdot 0 + a \cdot 0 \\ \implies 0 = a \cdot 0 &= a \cdot 1 = a \end{aligned}$$

□

Many theorems will state $1 \neq 0$ instead of $R \neq 0$.

(iii) $\mathbb{Z}$ with the usual multiplication and addition. Note that in $\mathbb{Z}/\{0\}$ we do not have a group with

respect to multiplication.

(iv) $\mathbb{Q}$ is a ring with the usual operations and $\mathbb{Q}/\{0\}$ is a group with respect to multiplication, that is $\mathbb{Q}$ is a field (multiplication in $\mathbb{Q}$ is commutative). $\mathbb{C}$ and $\mathbb{R}$ are fields as well.

(v) $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring with unity $\bar{1}$ ($\mathbb{Z}/n\mathbb{Z} = \{\bar{0}, \bar{1}, ..., n\bar{-}1\}$) where the multiplication is defined $\bar{a} \cdot \bar{b} = \bar{ab}$.

(vi) The quaternians: recall the imaginary units $i^2 = j^2 = k^2 = ijk = -1$. Looking at the set $\mathbb{H} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{R}\}$ where addition is defined by

$$(a + bi + cj + dk) + (a' + b'i + c'j + d'k) = (a + a') + (b + b')i + (c + c')j + (d + d')k$$

and multiplication is defined by distribution

$$\begin{aligned}&(a + bi + cj + dk)(a' + b'i + c'j + d'k)\\ =&aa' - bb' - cc' - dd' + (ab' + ba' + cd' - dc')i + (a'c - bd' + ca' + db')j + (ad' + bc' - cb' + da')k\end{aligned}$$

Then $\mathbb{H}$ forms a ring. We see that, for $x \in \mathbb{H}$,

$$\begin{aligned}x\bar{x} &= (a + bi + cj + dk)(a - bi - cj - dk) = a^2 + b^2 + c^2 + d^2\\ x^{-1} &= \frac{\bar{x}}{x\bar{x}} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}\end{aligned}$$

each analogous to the complex numbers. Every $x \neq 0$ in $\mathbb{H}$ has a multiplicative inverse. However, multiplication does not commute in all of $\mathbb{H}$ ($ik = -j \neq j = ki$), so $\mathbb{H}$ is a division ring but not a field.

(vii) Let X be a nonempty set and A be any ring. The set of all mappings $f : X \to A$ where $(f + g)(x) = f(x) + g(x)$ and $(fg)(x) = f(x)g(x)$ forms a ring.

Since rings add an additional operation to a group structure, we have new properties that arise from the interaction of the two operations.

Proposition 1.1.1. Let R be a ring.

(i) $0 \cdot a = a \cdot 0 = 0 \quad \forall a, b \in R$

(ii) $(-a)b = a(-b) = -(ab) \quad \forall a, b \in R$

(iii) $(-a)(-b) = ab \quad \forall a, b \in R$

(iv) If R has identity 1, then that identity is unique and

$$-a = (-1)a \quad \forall a \in R$$

Proof. (i) Given $a \in R$, we have

$$\begin{aligned}a \cdot 0 = a(0 + 0) &= a \cdot 0 + a \cdot 0\\ \implies 0 &= a \cdot 0\end{aligned}$$

Similarly,

$$\begin{aligned}0 \cdot a = (0 + 0)a &= 0 \cdot a + 0 \cdot a\\ \implies 0 &= 0 \cdot a\end{aligned}$$

(ii) Given $a, b \in R$, we have

$$\begin{aligned}ab + (-a)b &= (a + -a)b\\ &= 0 \cdot b\\ &= 0 \implies -(ab) = (-a)b\end{aligned}$$

$-(ab) = a(-b)$ is analogous.

(iii) Given $a, b \in R$, we have

$$\begin{aligned} -(ab) + (-a)(-b) &= (-a)b + (-a)(-b) \\ &= (-a)(b + -b) \\ &= (-a) \cdot 0 \\ &= 0 \\ \implies -(-(ab)) &= (-a)(-b) \\ \implies ab &= (-a)(-b) \end{aligned}$$

(iv) Let 1 and e both be identity elements in R. Then

$$\left.\begin{aligned} 1 \cdot e &= e \\ 1 \cdot e &= 1 \end{aligned}\right\} \implies 1 = e$$

Thus the identity element of R is unique. Let $a \in R$ be given. We have

$$\begin{aligned} 0 &= (1 + (-1))a \\ &= 1 \cdot a + (-1) \cdot a \\ &= a + (-1)a \\ \implies -a &= (-1)a \end{aligned}$$

□

1.2 Zero Divisors and Integral Domains

This semester we are adding multiplicative structure and we hope it is a good structure, but sometimes it is not. Consider the statement $3 \cdot 4 = 0$. Is this a true statement? In $\mathbb{R}$, no, it is not. However, if we consider the ring $\mathbb{Z}/6\mathbb{Z}$, then the statement is true. We call this blend of bad multiplicative behavior zero divisors.

Definition 1.2.1. (Zero Divisor)
Let R be a ring. A nonzero element $a \in R$ is called a zero divisor if there exists a nonzero $b \in R$ such that $ab = 0$ or $ba = 0$.

Example 1.2.1. Consider the matrices in $M_2(\mathbb{R})$. Let $A = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$
$$BA = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

We now introduce the notion of multiplicative inverses. For a ring to be field, every element needs to have a multiplicative inverse. However, in some rings we might have some elements that have inverses but not all. We call these elements units.

Definition 1.2.2. (Unit)
Assume a ring R has identity $1 \neq 0$. An element $a \in R$ is called a unit in R if there exists an element $u \in R$ such that $uv = 1 = vu$. The set of all units in R is denoted by $R^\times$.

Example 1.2.2. $\mathbb{Z}/6\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}\}$, $\mathbb{Z}/9\mathbb{Z} = \{\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}, \bar{6}, \bar{7}, \bar{8}\}$.
Here, $(\mathbb{Z}/6\mathbb{Z})^\times = \{\bar{1}, \bar{5}\}$ and $(\mathbb{Z}/9\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \bar{4}, \bar{5}, \bar{7}, \bar{8}\}$.

Concerning rings of the form $\mathbb{Z}/n\mathbb{Z}$ for $n \geq 0$, we have the following result regarding units.

Proposition 1.2.1. $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ is a unit if and only if $\gcd(a, n) = 1$.

Proof. ($\Longrightarrow$) Suppose $\bar{a}$ is a unit. Then there exists $\bar{b} \in R$ such that $\bar{a}\bar{b} = \bar{1}$. This means that $ab \equiv 1($ mod $n)$, so $n \mid ab - 1$. Thus there exists some $y \in \mathbb{Z}$ such that $ab - 1 = ny$. More specifically, $ab - ny = 1$. Thus any common divisor d of a and n must also divide 1, so $d = 1$. Hence $\gcd(a, n) = 1$.
($\Longleftarrow$) Suppose $\gcd(a, n) = 1$. By Bézout's identity, there exist $x, y \in \mathbb{Z}$ such that $ax + ny = 1$. Thus $ax \equiv 1($ mod $n)$, so $\bar{a}\bar{x} = \bar{1}$. Hence, $\bar{a}$ is a unit. □

Remark 1.2.1. A consequence of the above proposition: $\mathbb{Z}/p\mathbb{Z}$ where p is prime is a field as $(\mathbb{Z}/p\mathbb{Z})^t \; imes = \{\bar{1}, \bar{2}, ..., p - \bar{1}\}$.

Remark 1.2.2. Any nonzero element $\bar{a}$ of $\mathbb{Z}/n\mathbb{Z}$ with $\gcd(a, n) > 1$ is a zero divisor.

Example 1.2.3. In $Z/12\mathbb{Z}$, $\bar{8}$ is not relatively prime to 12 and $\bar{8}\bar{3} = \bar{2}\bar{4} = \bar{0}$.

In general, given $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ with $\bar{a} \neq 0$, we have that

$$\bar{a}\overline{\left(\frac{n}{\gcd(a, n)}\right)} = \frac{\bar{a}}{\gcd(a, n)} \cdot \bar{n} = \bar{0}$$

as $\frac{a}{\gcd(a,n)} \in \mathbb{Z}$. Further, since $\gcd(a, n) > 1$, we have that

$$0 < \frac{n}{\gcd(a, n)} < n$$

Proposition 1.2.2. Zero divisors can never be units.

Proof. Let R be a ring. Suppose that $a \in R$ is a unit and a zero divisor. Since a is a unit,

$$\exists b \in R \text{ such that } ab = 1 = ba$$

Since a is a zero divisor, "

$$\exists c \in R \text{ such that } c \neq 0 \text{ and } ac = 0 \text{ or } ca = 0$$

If $ac = 0$, then

$$\begin{aligned} & b(ac) = b \cdot 0 \\ \implies & (ba)c = 0 \\ \implies & 1 \cdot c = 0 \\ \implies & c = 0 \end{aligned}$$

a contradiction. The case for $ca = 0$ is analogous. □

It follows that fields contain no zero divisors. We now introduce the notion of an integral domain which is a ring with no zero divisors.

Definition 1.2.3. (Integral Domain)
A commutative ring with identity $1 \neq 0$ si called an integral domain if it has no zero divisors.

Proposition 1.2.3. Let R be a ring and assume $a, b, c \in R$ with a not a zero divisor. If $ab = ac$, then $b = c$ or $a = 0$.

Proof.

$$\begin{aligned} & ab = ac \\ \implies & ab - ac = 0 \\ \implies & a(b - c) = 0 \end{aligned}$$

a is not a zero divisor, so either $a = 0$ or $a \neq 0$. If $a = 0$, then we are done. If $a \neq 0$, then $b - c = 0$ which implies $b = c$. □

The proposition above applies for any integral domain. From this proposition, we can derive the following corollary.

Corollary 1.2.1. Any finite integral domain is a field.

Proof. Let R be a finite integral domain and let $a \in R$ such that $a \neq 0$. Define the mapping

$$\begin{aligned} & f_a : R \to R \\ & f_a(x) = ax \quad \forall x \in R \end{aligned}$$

By the previous proposition, f_a is injective and since R is finite f_a is a bijection. In particular,

$$\exists b \in R \text{ such that } f_a(b) = ab = 1$$

Therefore a is a unit and since a was an arbitrary nonzero element of R, R is a field. □

In group theory, an important structure of study was the subgroup. For rings, we have an analogous structure called a subring.

Definition 1.2.4. (Subring)
A subring of a ring R is a subgroup of R with respect to addition which is closed under multiplication.

Proposition 1.2.4. (Subring Check)
Let $H \subseteq R$ where R is a ring.

(i) Subgroup check:

  (a) Show H is nonempty (e.g., show $0 \in H$)

  (b) Show $\forall x, y \in H \quad x - y \in H$

(ii) Closure under multiplication: $\forall x, y \in H \quad xy \in H$

1.3 Polynomials and Ring Homomorphisms

One of the big motivations for studying rings is to understand polynomials. Polynomials are a fundamental object in algebra and in this section we will define ring homomorphisms to help us understand the structure of polynomials.

Definition 1.3.1. (Polynomial)
Let R be a ring with unity 1. Let x be an indeterminate.

(i) The sum
$$\sum_{i=0}^{n} a_i x^i = a_0 + a_1 x + a_2 x^2 + ... + a_n x^n = f(x)$$
with $n \geq 0$ and each $a_i \in R$ is called a polynomial in x with coefficients in R.

(ii) If $a_n \neq 0$, then $\deg(f(x)) = n$. The zero polynomial has all $a_i = 0$, and the degree of the zero polynomial is undefined. The zero polynomial is denoted by 0.

(iii) The collection of all polynomials with coefficients in R is denoted $R[x]$, and is a ring with respect to the opperations
$$\left(\sum_{i=0}^{n} a_i x^i\right) + \left(\sum_{i=0}^{n} b_i x^i\right) = \sum_{i=0}^{n} (a_i + b_i) x^i$$
and
$$\left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{i=0}^{n} b_i x^i\right) = a_0 b_0 + (a_0 b_1 + b_0 a_1)x + (a_0 b_2 + a_1 b_1 + a_2 b_0)x^2 + ... + (a_0 b_n + a_1 b_{n-1} + ... + a_n b_0)x^n$$

Example 1.3.1. When $R = \mathbb{Z}/3\mathbb{Z}$ (the polynomial ring $\mathbb{Z}/3\mathbb{Z}[x]$) Let $p(x) = x^2 + 2x + 1$ and $q(x) = x^3 + x + 2$. Then
$$p(x) + q(x) = x^3 + x^2 + 3x + 3 = x^3 + x^2$$
and
$$p(x)q(x) = x^5 + 2x^4 + 2x^3 + 4x^2 + 5x + 2$$

Example 1.3.2. In $\mathbb{Z}/2\mathbb{Z}[x]$
$$x^2 + 1 = (x + 1)(x + 1)$$

Ring homomorphisms are defined in a similar way to group homomorphisms.

Definition 1.3.2. (Ring Homomorphism)
Let R and S be rings.

(i) A ring homormorphism is a mapping $\phi : R \to S$ such that for all $a, b \in R$

(a) $\phi(a + b) = \phi(a) + \phi(b)$
(b) $\phi(ab) = \phi(a)\phi(b)$

(ii) The kernel of ϕ is defined as
$$\ker \phi = \{x \in R : \phi(x) = 0_S\}$$

(iii) A bijective ring homomorphism is an isomorphism.

Example 1.3.3. Let $\phi : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$ be defined by
$$\phi(x) = \begin{cases} 0 & \text{if } x \text{ is even} \\ 1 & \text{if } x \text{ is odd} \end{cases}$$

Case 1: x and y are both even.
$$\phi(x + y) = 0 = \phi(x) + \phi(y), \phi(xy) = 0 = 0 \cdot 0 = \phi(x)\phi(y)$$

Case 2: x and y are both odd.
$$\phi(x + y) = 0 = 1 + 1 = \phi(x) + \phi(y), \phi(xy) = 1 = 1 \cdot 1 = \phi(x)\phi(y)$$

Case 3: One of x and y is even and the other is odd.
Without loss of generality, let x be even and y be odd.

$$\phi(x+y) = 1 = 0+1 = \phi(x)+\phi(y), \phi(xy) = 0 = 0\cdot 1 = \phi(x)\phi(y)$$

We have two notable properties involving ring homomorphisms.

Proposition 1.3.1. Let R and S be rings and let $\phi : R \to S$ be a ring homomorphism.

(i) $\phi(R)$ is a subring of S.

(ii) $\ker\phi$ is a subring of R. Furthermore, if $\alpha \in \ker\phi$ and $r \in R$, the $r\alpha \in \ker\phi$ and $\alpha r \in \ker\phi$. That is, $\ker\phi$ is closed under the entire ring multiplication.

Proof. (i) We know that $\phi(R)$ is an additive subgroup of S from last semester. It remains to show that if $s_1, s_2 \in \phi(R)$, then $s_1 s_2 \in \phi(R)$. Let $s_1, s_2 \in \phi(R)$ be given. Then

$$\exists r_1, r_2 \in R \text{ such that } s_1 = \phi(r_1) \text{ and } s_2 = \phi(r_2)$$

Then

$$s_1 s_2 = \phi(r_1)\phi(r_2) = \phi(r_1 r_2) \in \phi(R)$$

as desired.

(ii) We know that $\ker\phi$ is an additive subgroup of R from last semester. It suffices to show that if $r \in R$ and $\alpha \in \ker\phi$, then $r\alpha, \alpha r \in \ker\phi$. Let $r \in R$ and $\alpha \in \ker\phi$ be given. We have

$$\begin{aligned}\phi(r\alpha) &= \phi(r)\phi(\alpha)\\ &= \phi(r)\cdot 0_S\\ &= 0_S\\ \implies r\alpha &\in \ker\phi\end{aligned}$$

Similarly,

$$\begin{aligned}\phi(\alpha r) &= \phi(\alpha)\phi(r)\\ &= 0_S \cdot \phi(r)\\ &= 0_S\\ \implies \alpha r &\in \ker\phi\end{aligned}$$

□

The kernel of a ring homomorphism is a special type of subring. In particular, the kernel of a ring homomorphism is closed under multiplication by any element in the ring. We call this the absorbative property it allows us to define coset multiplication for rings.

Let I be the kernel of a ring homomorphism $\phi : R \to S$. We know that the additive left cosets of I form a group. Because every subgroup of an abelian group is normal, the definition of coset addition is well-defined:

$$(\alpha + I) + (\beta + I) = (\alpha+\beta) + I \quad \forall \alpha, \beta \in R$$

We aim to show that the absorbative property of $\ker\phi = I$ makes coset multiplication well-defined. We define coset multiplication as

$$(\alpha + I)(\beta + I) = (\alpha\beta) + I$$

Suppose that $\alpha + I = \alpha' + I$ and $\beta + I = \beta' + I$. We want to show that

$$(\alpha\beta) + I = (\alpha'\beta') + I$$

Note that

$$\begin{aligned}\alpha + I = \alpha' + I &\implies \alpha = \alpha' + x \text{ where } x \in I\\ \beta + I = \beta' + I &\implies \beta = \beta' + y \text{ where } y \in I\end{aligned}$$

Then

$$\alpha\beta = (\alpha' + x)(\beta' + y) = \alpha'\beta' + \alpha' y + x\beta' + xy$$

$\alpha' y, x\beta', xy \in I$, so $\alpha\beta = \alpha'\beta' + z$ where $z = \alpha' y + x\beta' + xy \in I$. Thus $\alpha\beta - \alpha'\beta' \in I$. Therefore, $(\alpha\beta)+I = (\alpha'\beta')+I$ and so coset multiplication is well-defined. We have now motivated the following definition.

Definition 1.3.3. (Ideals)
Let R be a ring and let $I \subseteq R$. Let $r \in R$.

(i) $rI = \{rx : x \in I\}$ and $Ir = \{xr : x \in I\}$

(ii) I is a left ideal of R if

(a) I is a subring of R

(b) $\forall r \in R \;\; rI \subseteq I$

(iii) I is a right ideal of R if

(a) I is a subring of R

(b) $\forall r \in R \;\; Ir \subseteq I$

(iv) If I is both a left and right ideal of R, then I is called an ideal of R.

1.4 Quotient Rings and the Isomorphism Theorems

Now that we have defined ideals, we can use them to construct new rings called quotient rings, similar to how we used normal subgroups to construct quotient groups. We will then use ideals and quotient rings to discuss isomorphisms between ring structures. First, given a ring R and an ideal I of R, we will show that the set R/I has any meaning.

Proposition 1.4.1. Let R be a ring and I be an ideal of R. Then R/I is a ring under the binary operations

$$\forall r, s \in I \quad (r+I)+(s+I) = (r+s)+I$$
$$\forall r, s \in I \quad (r+I)(s+I) = (rs)+I$$

Proof. Note that we proved multiplication is well-defined and we also know the group structure exists from last semester. Distribution and associativity in R/I follow from distribution and associativity in R. □

Definition 1.4.1. (Quotient Ring)
When I is an ideal of a ring R, the ring R/I is called the quotient ring of R by I.

We can now establish the isormophism theorems for rings.

Theorem 1.4.1. (The First Isomorphism Theorem for Rings)

(i) If $\phi : R \to S$ is a ring homomorphism, then $\ker\phi$ is an ideal of R, $\phi(R)$ is a subring of S, and

$$\frac{R}{\ker\phi} \cong \phi(R)$$

(ii) If I is any ideal of R, then the map

$$\pi : R \to R/I$$
$$\pi(r) = r + I$$

is a surjective ring homomorphism with kernel I. This is called the natural projection of R onto R/I.

Proof. Last semester we established the mapping

$$\mu : R/\ker\phi \to \phi(R)$$
$$\mu(r+K) = \phi(r)$$

as a well-defined bijection which preserved the group operation (in this case, $+$). It remains to show that μ preserves coset multiplication.

$$\begin{aligned}\mu\left((r+K)(s+K)\right) &= \mu(rs+K)\\ &= \phi(rs)\\ &= \phi(r)\phi(s)\\ &= \mu(r+K)\mu(s+K)\end{aligned}$$

□

Example 1.4.1. Let R be a ring.

(i) The trivial ideal is 0.

(ii) R is an ideal of R.

(iii) If I is an ideal of R where $I \neq R$, then I is a proper ideal of R.

Example 1.4.2. Last semester (when studying cyclic groups) we proved the only subgroups of $\mathbb{Z}$ were of the form $n\mathbb{Z}$ for some $n \geq 0$. This tells us that all the ideals of $\mathbb{Z}$ must be of the same form. It is a

straightforward exercise to show that $n\mathbb{Z}$ $(n \geq 0)$ is an ideal of $\mathbb{Z}$:

$$x \in \mathbb{Z}, y \in n\mathbb{Z} \implies y = nz \quad \text{for } z \in \mathbb{Z} \text{ and } xy = x(nz) = n(xz) \in n\mathbb{Z}$$
$$yx = (nz)x = n(zx) \in n\mathbb{Z}$$

In this setting,

$$\pi : \mathbb{Z} \to \mathbb{Z}/n\mathbb{Z}$$
$$\pi(x) = \bar{x} \text{ (reduction modulo } n)$$
$$\mathbb{Z}/5\mathbb{Z} \quad 7 = 2$$

Example 1.4.3. Consider $I = \{p(x) \in \mathbb{Z}[x] : p(x) = 0 \text{ or whose terms are of at least degree } 2\}$. Does I have the absorbative property of an ideal? Let $f(x) \in \mathbb{Z}[x]$ and $q(x) \in I$. Then

$$f(x) = \sum_{i=0}^{n} a_i x^i \quad q(x) = \sum_{i=0}^{n} b_i x^i \quad \text{where } b_0 = 0, b_1 = 0$$

Then the constant coefficient of $f(x)q(x)$ is $a_0 b_0 = a_0 \cdot 0 = 0$ and the linear coefficient of $f(x)q(x)$ is $a_1 b_0 + a_0 b_1 = a_1 \cdot 0 + a_0 \cdot 0 = 0$.

Example 1.4.4. Let A be a ring and X be a nonempty set. Let R be the ring of all functions from X to A. For each $c \in X$ we can define the mapping

$$E_c : R \to A$$
$$E_c(f) = f(c) \text{ (the evaluation homomorphism)}$$

Let $f, g \in R$. Then

$$\begin{aligned} E_c\left((f \cdot g)(x)\right) &= (f \cdot g)(c) \\ &= f(c) \cdot g(c) \\ &= E_c\left(f(x)\right) \cdot E_c\left(g(x)\right) \end{aligned}$$

Addition follows similarly. Notice

$$\ker E_c = \{f \in R : E_c(f) = 0\} = \{f \in R : f(c) = 0\}$$

By the first isomorphism theorem, we have

$$\frac{R}{\ker E_c} \cong E_c(R)$$

What is $E_c(R)$? Let $a \in A$. Notice the constant function $h(x) = a \ \forall x \in X$ gives

$$E_c\left(h(x)\right) = h(c) = a$$

Thus $E_c(R)$ is surjective and so

$$\frac{R}{\ker E_c} \cong A$$

Theorem 1.4.2. (The Second, Third, and Fourth Isomorphism Theorems for Rings)
Let R be a ring.

(i) Let A be a subring of R and let B be an ideal of R. Then

$$A + B = \{a + b : a \in A, b \in B\}$$

is a subring of R, $A \cap B$ is an ideal of A, and

$$\frac{A + B}{B} \cong \frac{A}{A \cap B}$$

(ii) Let I and J be ideals of R with $I \subseteq J$. Then J/I is an ideal of R/I and

$$\frac{R/I}{J/I} \cong \frac{R}{J}$$

(iii) Let I be an ideal of R. The correpsondence $A \leftrightarrow A/I$ is an inclusion preserving bijection between the set of subrings of R containing I and the subrings of R/I. Furthermore, A (a subring containing I) is an ideal of R if and only if A/I is an ideal of R/I.

Similar to groups, ring homomorphisms will map subrings to subrings. Can the same be said for ideals? The answer is no, this is only true for surjective homomorphisms.

1.5 Generated Ideals

Proposition 1.5.1. Let I and J be ideals of a ring R. Define

$$I + J = \{a + b : a \in I, b \in J\}$$

Then $I + J$ is the smalles ideal of R that contains both I and J.

Proof. We know that I and J are normal additive subgroups of R, so $I + J$ is a subgroup of R.

$$\begin{aligned} a_1 + b_1, a_2 + b_2 \in I + J \implies (a_1 + b_1)(a_2 + b_2) \\ = a_1a_2 + a_1b_2b_1a_2 + b_1b_2 \end{aligned}$$

Notice

$$\begin{aligned} a_1 \in I \implies \begin{matrix} a_1a_2 \in I \\ a_1b_2 \in I \end{matrix} \end{aligned} \qquad \begin{aligned} b_1 \in J \implies \begin{matrix} b_1b_2 \in J \\ b_1a_2 \in J \end{matrix} \end{aligned}$$

since I and J are ideals. Hence, $(a_1 + b_1)(a_2 + b_2) \in I + J$ and so $I + J$ is an ideal of R.
Futhermore, $I + J$ is the smalles ideal of R which contains I and J. We can verify that I and J are contained in $I + J$. Letting $b = 0$, we see any $a \in I$ is in $I + J$. Similarly, letting $a = 0$ we see any $b \in J$ is in $I + J$. Hence both I and J are contained in $I + J$. Let K be an ideal of R which contains both I and J. Let $a + b \in I + J$. Then

$$\begin{aligned} a \in I \text{ and } b \in J &\implies a \in K \text{ and } b \in K \\ &\implies a + b \in K \\ &\implies I + J \subseteq K \end{aligned}$$

□

Definition 1.5.1. (Finitely Generated Ideal)
Let R be a ring with identity 1. Let A be a subset of R. Let (A) denote the smallest ideal of R containing A. We can define (A) more concretely by

$$(A) = \bigcap_{I \subseteq A} I, \ I \text{ is an ideal of } R$$

An ideal genearted by a finite set is called a finitely generated ideal. If $A = \{a_1, a_2, ..., a_n\}$, then (A) will be written as $(a_1, a_2, ..., a_n)$.
$(I, J) = I \cup J$ (where I and J are subsets of R)

What is the simplest group generated by a set? Cyclic groups that are generated with a single element. Given a ring R with unity $1 \neq 0$ and some $a \in R$, consider (a), the ideal generated by a. (a) needs to contain ra and ar for all $r \in R$. What's more is (a) must be closed with respect to addition so we need all finite sums of ra's and ar's

$$(r_1a + r_2a + ar_3 + ar_4 + ...)$$

If we suppose that R is commutative, then (a) is very simple.

Definition 1.5.2. (Principal Ideal)
Let R be a ring and $a \in R$. When R is commutative with $1 \neq 0$,

$$(a) = \{ar : r \in R\}$$

and (a) is called the principal ideal of R generated by a.

Example 1.5.1. (i) In any ring, $0 = (0)$ and when R has unity 1, $(1) = R$.

(ii) Let R be a commutatie ring with unity $1 \neq 0$.

$$(a) + (b) = (a + b) \text{ (from what we proved about } I + J)$$

(iii) All ideals are principal.

(iv) All additive subgroups of $\mathbb{Z}$ are of the form $n\mathbb{Z}, n \geq 0$ and $n\mathbb{Z}$ is an ideal for all $n \geq 0$. Furthermore, $n\mathbb{Z} = \{nx : x \in \mathbb{Z}\} = (n)$.

Proposition 1.5.2. Let I be an ideal of R.

(i) $I = R$ if and only if I contains a unit.

(ii) Assume R is commutative. R is a field if and only if its only ideals are 0 and R.

Proof. (i) If $I = R$, then $1 \in I$. Conversely, if u is a unit in I with inverse $v \in R$, then for any $r \in R$

$$r = r \cdot 1 = r(vu) = (rv)u \in I$$

Thus $R = I$.

(ii) The ring R is a field if and only if every nonzero element is a unit. Hence, any nonzero ideal contains a unit. Thus the only nonzero ideal of R is R. Conversely, if 0 and R are the only ideals of R, we let u be any nonzero element of R. By hypothesis, $(u) = R$ so $1 \in (u)$. That is, $1 = uv$ for some $v \in R$. That is, u is a unit. Thus every nonzero element of R is a unit so R is a field.

□

Let's say I is a nonzero ideal in $\mathbb{Q}$. Then there exists $a \in I$ such that $a \neq 0$. Since ideals have the absorbative property, we know that $\frac{1}{a} \in \mathbb{Q}$ and $\frac{1}{a} \cdot a \in I$ which implies $1 \in I$. Then for any $b \in \mathbb{Q}$ we have $b = b \cdot 1 \in I$ which implies $\mathbb{Q} \subseteq I$. Thus $I = \mathbb{Q}$.

1.6 Maximal Ideals

Note that for any $n \in \mathbb{N}$ and $a \in R$ a ring, we define

$$na = a + a + ... + a \quad n \text{ times}$$

Proposition 1.6.1. Let $m, n \in \mathbb{N}$ such that $\gcd(m, n) = 1$. Then $m \mid \binom{m}{n}$.

Proof.

$$\begin{aligned}\binom{m}{n} &= \frac{m!}{n!(m-n)!} \\ &= \frac{m}{n}\frac{(m-1)!}{(n-1)!(m-n)!} \\ &= \frac{m}{n}\binom{m-1}{n-1}\end{aligned}$$

So,

$$n\binom{m}{n} = m\binom{m-1}{n-1}$$

so $m \mid n\binom{m}{n}$. But $\gcd(m, n) = 1$, so $m \mid \binom{m}{n}$. □

Corollary 1.6.1. If R is a field, then any nonzero ring homomorphism from R into another ring is an injection.

Proof. Let R be a field and let $\phi : R \to S$ be a nonzero ring homomorphism. Since $\ker \phi$ is an ideal fo the field R, $\ker \phi$ is either 0 or R itself. If $\ker \phi = R$, then $\phi(r) = 0$ for all $r \in R$. Hence, contrary to our assumption, ϕ is the zero homomorphism. This leaves us with $\ker \phi = 0$. As we know from our work with groups,

$$\ker \phi = 0 \iff \phi \text{ is injective}$$

□

Definition 1.6.1. (Maximal Ideal)
An ideal M in a ring S is called a maximal ideal if $M \neq S$ ans the only ideals that contain M are S and M itself.

A maximal ideal gives the notion of a "largest proper ideal" of a ring. That is, if M is the maximal ideal to a ring R and H is a proper ideal of R such that $M \subseteq H \subseteq R$, then either $H = R$ or $H = M$.

Proposition 1.6.2. In a ring with unity, every proper ideal is contained in a maximal ideal.

The next result characterizes maximal ideals with their quotient structures of a commutative ring.

Proposition 1.6.3. Assume R is commutative. The ideal M is a maximal ideal if and only if R/M is a field.

Proof. M is a maximal ideal if and only if there are no ideals I with

$$M \subset I \subset R$$

By the fourth isomorphism theorem, ideals of R containing M are in 1-1 correspondence with the ideals of R/M. So M is maximal if and only if there are no ideals I with $M \subset I \subset R$ if and only if there are no maximal ideals of R/M if and only if the only ideals of R/M are R/M and 0 if and only if R/M is a field. Some things to verify: R/M is commutative (since R is commutative, we have R/M is commutative), R/M has unity ($1 + M \in R/M$), and R/M is not the zero ring ($M \neq R, R/M \neq 0$). □

Example 1.6.1. What are the maximal ideals in $\mathbb{Z}$? We know all subrings of Z are of the form $n\mathbb{Z}, n \geq 0$. It takes very little work to show that $n\mathbb{Z}, n \geq 0$ is an ideal. The maximal ideals are those such that $\mathbb{Z}/n\mathbb{Z}$ is field. We know that $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime. Thus $n\mathbb{Z}, n \geq 0$ is a maximal ideal if and only if n is prime.

We now look to generalize the notion of a prime number through ideals. Looking at $\mathbb{Z}$, suppose $n\mathbb{Z}, n \geq 0$ is an ideal of $\mathbb{Z}$. Suppose $a, b \in \mathbb{Z}$ such that $a \in n\mathbb{Z}$ or $b \in n\mathbb{Z}$. Note

$$a \in (n) \iff a = nx \quad \text{for some integer } x$$

So $n \mid a$. Equivalently, we can say if $n \mid ab$ then $n \mid a$ or $n \mid b$. This is exactly the behavior of a prime number. This motivates the following definition.

Definition 1.6.2. (Prime Ideal)
Assume R is commutative. An ideal P is called a prime ideal if $P \neq R$ and if $ab \in P$, then $a \in P$ or $b \in P$.

The defining trait of a prime ideal is that whenever a product lands in the ideal, one of the factors must already be there. In a quotient ring, this looks like the following: if $(a + P)(b + P) = 0 + P$ then either $(a + P) = 0 + P$ or $(b + P) = 0 + P$. This leads us to the following characterization.

Proposition 1.6.4. Assume R is commutative. The ideal P is a prime ideal in R if and only if R/P is an integral domain.

Proof. The ideal P is prime if and only if $P \neq R$ and if $ab \in P$, then $a \in P$ or $b \in P$. Recall that

(i) $ab \in P \iff (a + P)(b + P) = ab + P = 0 + P$

(ii) $a \in P \iff a + P = 0 + P$

(iii) $b \in P \iff b + P = 0 + P$

So, P is a prime ideal if and only if $R/P \neq 0$ and if $(a + P)(b + P) = 0 + P$, then $a + P = 0$ or $b + P = 0$ if and only if R/P is an integral domain (since we know R/P is a commutative ring with unity). □

1.7 Rings of Fractions

We take a brief look at fractions. Fractions are equivalence cleases, as given any two fractions a/b and c/d we have

$$\frac{a}{b} = \frac{c}{d} \iff da = cb$$

If we look at $R \times R$ where R is a ring we can define fractions as ordered pairs under the relation

$$(a, b) = (c, d) \iff da = cb$$

What this allows us to do when R is an integral domain is to create a field which contains an isomorphic copy of R. We will not spend much time here, so we end this chapter with the following remark.

Remark 1.7.1. Every integral domain is contained in a field and the smalles field which contains an integral domain is called the field of fractions of the integral domain.

In the next chapter, we will discuss different types of integral domains. More specifically, we will discuss the following:

(i) Euclidean domain (ED)

(ii) Principal ideal domain (PID)

(iii) Unique factorization domains (UFD)

Chapter 2

Euclidean Domains, Principal Ideal Domains, and Unique Factorization Domains

2.1 Euclidean Domains

Having seen how prime ideals generalize the idea of prime numbers to ring structures, and their relation to integral domains, we now focus more closely on the arithmetic of integral domains. In number theory, division with remainders and unique factorization are tightly linked concepts. We wish to generalize these concepts to integral domains, capturing when an integral domain supports a robust theory of greatest common divisors and factorizations.

We first start with integral domains that have a sense of a division algorithm. We will call these Euclidean Domains. First, here are some examples where a division algorithm exists.

Example 2.1.1. (i) $\mathbb{Z}$ is a euclidean domain.

(ii) $F[x]$, the set of polynomials over a field, is a euclidean domain.

(iii) Any field will be a euclidean domain.

(iv) The Gaussian integers, $\{a + bi : a, b \in \mathbb{Z}\}$, is a euclidean domain.

Before we formally define a euclidean domain, we first need to define special functions on a ring that would allow us to terminate an algorithm. More specifically, if we wish to have a division algorithm, we would need a way to specifiy that the remainder becomes provably smaller. This avoids the issue of having repeated remainders that continue forever.

Definition 2.1.1. (Norm on a Ring)
Any function $N : R \to \mathbb{N} \cup \{0\}$ (where R is a ring) with $N(0) = 0$ is called a norm on the ring R. If $N(a) > 0$ for all $a \in R, a \neq 0$, we call N a positive norm.

We now define euclidean domains.

Definition 2.1.2. (Euclidean Domain)
An integral domain R is said to be a euclidean domain (ED) if there exists a norm on R such that for any $a, b \in R$ with $b \neq 0$ there exists $q, r \in R$ such that

$$a = bq + r$$

where $r = 0$ or $N(r) < N(b)$.

A division algorithm gives us a euclidean algorithm. This algorithm is repeated division of some a and b in

the euclidean domain. Given a euclidean domain R and $a, b \in R$ with $b \neq 0$,

$$\begin{aligned} a &= q_0 b + r_0 \text{ where } r_0 = 0 \text{ or } N(r_0) < N(b) \\ b &= q_1 r_0 + r_1 \text{ where } r_1 = 0 \text{ or } N(r_1) < N(r_0) \\ r_0 &= q_2 r_1 + r_2 \text{ where } r_2 = 0 \text{ or } N(r_2) < N(r_1) \\ &\vdots \\ r_{n-2} &= q_n r_{n-1} + r_n \text{ where } r_n = 0 \text{ or } N(r_n) < N(r_{n-1}) \end{aligned}$$

Because the codomain of the norm is $\mathbb{N} \cup \{0\}$ we have created a strictly decreasing sequence so it must terminate with some remainder of zero.

$$N(b) > N(r_0) > N(r_1) > ... > N(r_{n-1}) > N(r_n)$$

Example 2.1.2. Some examples of norms.

(i) $\mathbb{Z}$, $N(x) = |x|$

(ii) $F[x]$, $N(f(x)) = \deg(f(x))$

(iii) For any field $\mathbb{F}$, any norm works: $N(x) = 0 \ \forall x \in \mathbb{F}$, $N(x) = 69 \ \forall x \in \mathbb{F}$, etc. There are no remainders when dividing in a field, so there is no need for a norm to terminate the euclidean algorithm.

(iv) $\mathbb{G} = \{a + bi : a, b \in \mathbb{Z}\}$ (the Gaussian integers), $N(a + bi) = a^2 + b^2$

Proposition 2.1.1. Every ideal in a euclidean domain is a principal ideal. More precisely, if I is any nonzero ideal in the euclidean domain R then $I = (d)$ where d is any nonzero element of I with minimum norm.

Proof. If I is the zero ideal, then $I = (0)$. Suppose $I \neq 0$. Then let d be a nonzero element of I with minimal norm (such a d exists since the set $\{N(a) : A \in I\}$ is a nonempty set of integers bounded below and by the well-ordering principal there must be a minimal norm attained by an element of I). Clearly, $(d) \subseteq I$ since $d \in I$. To show $I \subseteq (d)$ let $a \in I$ and use the division algorithm to write

$$a = qd + r$$

where $q, r \in R$ and $r = 0$ or $N(r) < N(d)$. Then

$$r = a - qd$$

and since I is an ideal and $q \in R$ with $d \in I$, we have $qd \in I$. Further, $a \in I$ and so $a - qd \in I$. Thus $r \in I$. Note that $N(d) < N(r)$, which contradicts minimality of the norm of d. Hence, the only option is that $r = 0$. Hence

$$a = qd \implies a \in (d)$$

Therefore, $I \subseteq (d)$. It follows that $I = (d)$. □

2.2 Number Theory in a Ring

Now that we have a notion of a euclidean domains, we proceed to establish arithmetic on such rings, notably divisibility and greatest common divisors.

Definition 2.2.1. (Divide, Greatest Common Divisor)
Let R be a commutative ring and let $a, b \in R$ with $b \neq 0$. The element b is said to divide a if

$$\exists r \in R \text{ such that } br = a$$

If b divides a, we write $b \mid a$.
The greatest common divisor (GCD) of a and b is a nonzero element of R, d, such that

(i) $d \mid a$ and $d \mid b$

(ii) If $d' \mid a$ and $d' \mid b$, then $d' \mid d$

The notation remains the same:

$$d = \gcd(a, b) \text{ or } d = (a, b)$$

In the language of ideals if I is the ideal generated by (a, b) then d is a $\gcd(a, b)$ if

(i) $I \subseteq (d)$

(ii) If $I \subseteq (d')$ then $(d) \subseteq (d')$

Proposition 2.2.1. If a and b are nonzero elements of a commutative ring R such that the ideal generated by a and b is a principal ideal (d), then d is a greatest common divisor of a and b.

Proof. Suppose $I = (a, b) = (d)$. Then

$$a, b \in (d) \implies d \mid a \text{ and } d \mid b$$

Let $d' \in R$ such that $(a, b) \subseteq (d')$. Then $a \in (d')$ and $b \in (d')$ implies $d' \mid a$ and $d' \mid b$. Since $(d) = (a, b)$, $(d) \subseteq (d')$ which implies $d \in (d')$. More specifically, $d' \mid d$. Hence, d is a greatest common divisor of a and b. □

Just like for standard integers, the greatest common divisor is unique.

Proposition 2.2.2. Let R be an integral domain. If d and d' in R both generate the same principal ideal, i.e. $(d) = (d')$, then $d = ud'$ where u is a unit in R. In particular, if d and d' are both greatest common divisors of a and b, then $d' = wd$ where w is some unit in R.

Proof. This is trivial if d or d' is zero. Assume d and d' are both nonzero. Since $(d) = (d')$, we have that $d' \in (d)$ and so $\exists x \in R$ such that $dx = d'$. Since $d \in (d')$, $\exists y \in R$ such that $d'y = d$. This gives

$$\begin{aligned} d'yx = d' &\implies d'yx - d' = 0 \\ &\implies d'(yx - 1) = 0 \end{aligned}$$

Since $d' \neq 0$ and R is an integral domain, we know that $yx - 1 = 0 \implies yx = 1$. Hence, both x and y are units. □

2.3 Principal Ideal Domains

Previously, we have shown that every ideal in a euclidean domain is principal. Euclidean domains form a special case of what are called principal ideal domains. We now work in the more general principal ideal setting; in particular, each result for princiapl ideal domains that we prove will hold in any euclidean domain.

Definition 2.3.1. (Principal Ideal Domain)
A principal ideal domain (PID) is an integral domain where every ideal is principal.

Proposition 2.3.1. Let R be a principal ideal domain and let $a, b \in R$ where both a and b are nonzero. Let d be the generator of the ideal (a, b). Then

(i) d is a greatest common divisor of a and b

(ii) d can be written as an R linear combination of a and b: $\exists x, y \in R$ such that $ax + by = d$

(iii) d is unique up to unit multiples

In a commutative ring,

$$(d) = (a, b) = (a) + (b) = \{ax + by : x, y \in R\} \implies \exists x, y \in R \text{ such that } ax + by = d$$

Proposition 2.3.2. Every nonzero prime ideal in a principal ideal domain is maximal.

Proof. Let (p) be a nonzero prime ideal of a principal ideal domain R. Let I be an ideal that contains (p). Since R is a principal ideal domain, there exists some $i \in R$ such that $(i) = I$. Our goal is to show $I = (p)$ or $I = R$. Since $p \in (i)$, we have

$$p = ri \text{ for some } r \in R$$

So $ri \in (p)$. Since (p) is prime, $r \in (p)$ or $i \in (p)$. If $i \in (p)$ then $(i) \subseteq (p)$. Since $I = (i)$ was assumed to contain (p), we have $I = (p)$.
If $r \in (p)$, then $r = px$ for some $x \in R$. Then

$$p = ri \implies r = (ri)x$$

Since r is nonzero (ir $r = 0$ then $p = 0$, a contradiction), we have

$$1 = ix$$

Thus i and x are units. Since $i \in I$ we have that $I = R$. □

Fact 2.3.1. Here are some facts about $R[x]$ where R is a commutative ring:

(i) If R is an integral domain, $R[x]$ is an integral domain

(ii) R is contained in $R[x]$ where R is the set of all constant polynomials

Proposition 2.3.3. If R is any commutative ring such that the polynomial ring $R[x]$ is a principal ideal domain, then R is a field.

Proof. Assume $R[x]$ is a principal ideal domain. Since R is a subring of $R[x]$, then R must be an integral domain. The ideal (x) is a nonzero prime ideal in $R[x]$ because

$$\frac{R[x]}{(x)} \cong R \qquad (*)$$

is an ideal. Thus (x) is maximal, so $R[x]/(x)$ is a field. That is, R is a field.

Proof of $(*)$: Let $\sum_{i=0}^{n} a_i x^i \in R[x]$. Then in $R[x]/(x)$, we have

$$\sum_{i=0}^{n} a_i x^i + (x) = a_0 + (x)$$

Let's define a mapping

$$\psi : R[x] \to R$$
$$\psi(f) = f(0)$$

Let $f(x), g(x) \in R[x]$ where $f(x)$ has constant coefficient a_0 and $g(x)$ has constant coefficient b_0. Then

$$\psi(f(x)g(x)) = \psi(h(x)) = h(0) = a_0 b_0 = f(0)g(0) = \psi(f(x))\psi(g(x))$$

where $h(x) = f(x)g(x)$. Similarly,

$$\psi(f(x) + g(x)) = \psi(k(x)) = k(0) = a_0 + b_0 = f(0) + g(0) = \psi(f(x)) + \psi(g(x))$$

where $k(x) = f(x) + g(x)$. If $f(x) \in \ker \psi$, the $f(0) = 0$. That is, $f(x)$ has a zero constant coefficient. That is,

$$f(x) = a_1 x + a_2 x^2 + ... + a_n x^n = x(a_1 + a_2 x + ... + a_n x^{n-1})$$

so $f(x) \in (x)$. Note that if $p(x) \in (x)$, then $p(x)$ has zero constant coefficient so $p(0) = 0$ and $p(x) \in \ker \psi$. By the first isomorphism theorem,

$$\frac{R[x]}{\ker \psi} \cong \psi(R[x])$$
$$\implies \frac{R[x]}{(x)} \cong \psi(R[x])$$

Note ψ is onto since for any $r \in R$, $\psi(r) = r$ where we consider $r \in R[x]$ to be a constant polynomial. Hence

$$\frac{R[x]}{(x)} \cong R$$

□

2.4 Unique Factorization Domains

When finding the greatest common divisor for integers, we use the fundamental theorem of arithmetic to find prime factorizations. For example, if we have $m = 3^7 \cdot 2^3 \cdot 5^6$ and $n = 7^7 \cdot 2^5 \cdot 3^2$ then the greatest commond divisor is

$$\gcd(m, n) = 3^{\min\{2,7\}} \cdot 2^{\min\{3,5\}} = 3^2 \cdot 2^3$$

We would to extend this mechanism to rings.

Definition 2.4.1. (Reducible, Prime, Associate)
Let R be an integral domain.

(i) Suppose $r \in R$ is nonzero and not a unit. Then r is called irreducible if whenever $r = ab$ where $a, b \in R$, then one of a and b is a unit. Otherwise, r is said to be reducible.

(ii) A nonzero element $p \in R$ is called prime if (p) is a prime ideal (if $ab \in (p)$, then $a \in (p)$ or $b \in (p)$).

(iii) Two elements a and b of R differing by a unit multiple are called associate in R (i.e., $a = ub$ where $u \in R$ is a unit).

Proposition 2.4.1. In an integral domain, a prime is irreducible.

Proof. Suppose p is a prime in an integral domain R. So (p) is a nonzero prime ideal. Let $p = ab$ where $a, b \in R$. Then $ab \in (p)$. Since (p) is prime, $a \in (p)$ or $b \in (p)$. Without loss of generality, suppose $a \in (p)$. Then $a = pr$ where $r \in R$. Then

$$\begin{aligned} & p = (pr)b \\ \implies & 1 = rb \\ \implies & b \text{ is a unit in } R. \end{aligned}$$

Hence p is irreducible in R. □

In an integral domain, a prime element is irreducible. However, not every irreducible element is prime. For example $3 \in \left\{a + b\sqrt{-5} : a, b \in \mathbb{Z}\right\}$ is irreducible but not prime. Now if we consider a principal ideal domain, then we have the following result.

Proposition 2.4.2. In a principal ideal domain, a nonzero element is prime if and only if it is irreducible.

Proof. Let R be a principal ideal domain. Since principal ideal domains are integral domains, it automatically follows that any prime $p \in R$ is irreducible in R.
Let $p \in R$ be irreducible. Consider the ideal generated (p). Let M be an ideal such that $(p) \subseteq M$. Since R is a principal ideal domain, there exists $m \in R$ such that $M = (m)$. So $p \in (m)$ and thus $p = mx$ for some $x \in R$. Since p is irreducible, x or m must be a unit. If x is a unit, then m and p are associate, hence $(m) = (p)$. If m is a unit, then $(m) = R$. Hence (p) is maximal, which implies (p) is prime. □

Principal ideal domains provide a nice characterization for a prime involving reducibility. This mirrors how primes behave in the integers: a prime $p \in \mathbb{N}$, $p > 1$ is prime if its only factors are 1 and p (p is irreducible in $\mathbb{N}$). We now classify rings where factorizations are unique, allowing us to have a notion of the fundamental theorem of arithmetic.

Definition 2.4.2. (Unique Factorization Domain)
A unique factorization domain (UFD) is an integral domain R in which every nonzero $r \in R$ that is not a unit has the following properties:

(i) r can be written as a finite product of irreducibles p_i of R (not necessarily distinct)

$$r = p_1 p_2 \cdots p_n$$

(ii) The decomposition in part (i) is unique up to associates, namely if $r = q_1 q_2 \cdots q_m$ is another factorization of r, then $m = n$ and there is some reording of the factors such that $p_i = u_i q_i$ for all i where u_i is a unit in R.

Proposition 2.4.3. In a unique factorization domain a nonzero element is prime if and only if it is irreducible.

Proof. We only need to prove ($\Longleftarrow$). To that end, let p be an irreducible element in R and assume $p \mid ab$ for some $a, b \in R$. We will show that $p \mid a$ or $p \mid b$. Since $p \mid ab$, there exists some $d \in R$ such that $pd = ab$. Since R is a unique factorization domain, a and b can be written uniquely (up to assoicates) a product of irreducibles in R. That is,

$$a = p_1 p_2 \cdots p_n$$
$$b = q_1 q_2 \cdots q_m$$

where the p_i and q_i are irreducibles. Hence

$$ab = (p_1 p_2 \cdots p_n)(q_1 q_2 \cdots q_m)$$

which gives a unique (up to associates) factorization of ab. Thus

$$pd = p_1 p_2 \cdots p_n q_1 q_2 \cdots q_m$$

By the uniqueness of factorizations we know that p is an associate of one of the irreducibles on the right-hand side. If p is an associate of a p_i for some i, then $p \mid a$. Similarly, if p is an associate of a q_j for some j, then $p \mid b$. Thus p is prime. □

At this point, we have introduced three types of integral domains. We conclude this section by showing that every principal ideal domain is in fact a unique factorization domain. First, we need the following lemma.

Lemma 2.4.1. In a principal ideal domain R, given an ascending chain of ideals

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq ... \subseteq R$$

it eventually becomes stationary. That is,

$$\exists n \in \mathbb{N} \text{ such that } I_n = I_k \forall \;\; k \geq n$$

Proof. Let $I = \bigcup_{i=1}^{\infty} I_i$. It is easy to show I is an ideal. Since R is a principal ideal domain,

$$\exists p \in R \text{ such that } I = (p)$$

Since $p \in I = \bigcup_{i=1}^{\infty} I_i$, $p \in I_n$ for some $n \in \mathbb{N}$. Thus $(p) \subseteq I_n$ and by definition of unions $I_n \subseteq I = (p)$. Thus $I_n = (p)$ and for any $k \geq n$, we have that $I_k \supseteq I_n \supseteq I$ so it follows that $I_k = I$. □

Theorem 2.4.1. Every principal ideal domain is a unique factorization domain.

Proof. Let $r \in R$ where R is a principal ideal domain. Further assume that r is nonzero and not a unit. If r is irreducible, then we're done. If not, r is reducible so

$$r = r_1 r_2 \text{ for } r_1, r_2 \in R$$

where r_1 and r_2 are not units. If both r_1 and r_2 are reducible, then we're done. Otherwise, r_1 and r_2 are reducible. Supposing r_1 is reducible,

$$r_1 = r_{11} r_{12} \text{ for } r_{11}, r_{12} \in R$$

and so forth. If this process does not terminate, we obtain a proper inclusion of ideals. That is,

$$(r) \subset (r_1) \subset (r_{11}) \subset ...$$

where (r) is prperly in (r_1) since r_2 is not a unit and hence r and r_1 are associates. The rest of the proper containment follows similarly. However, in a principal ideal domain any ascending chain of ideals become stable so this contradicts the proper containment. Hence, we can factor r into a product of irreducible elements of R. □

To sumarize this chapter: every euclidean domain is a principal ideal domain, and every principal ideal domain

is a unique factorization domain.

$$\text{Euclidean domains} \subseteq \text{principal ideal domains} \subseteq \text{unique factorization domains}$$

Chapter 3

Polynomials and Vector Spaces

3.1 Polynomials Over a Field

Now that we have a foundation for factorization in a ring setting, we shift our focus to polynomial rings.

Recall . Let R be an integral domain. Then

(i) $\forall p, q \in R[x]$ with $p \neq 0$ and $q \neq 0$, $\deg(p(x)q(x)) = \deg(p(x)) + \deg(q(x))$.

(ii) The units in $R[x]$ are just units of R.

(iii) $R[x]$ is an integral domain.

If $R[x]$ is an integral domain, the field of fractions of $R[x]$ is

$$R(x) = \left\{ \frac{p(x)}{q(x)} : p, q \in R[x], q(x) \neq 0 \right\}$$

Proposition 3.1.1. Let I be an ideal of R and let $(I) = I[x]$ (the ideal generated by I in $R[x]$). Then

$$\frac{R[x]}{(I)} \cong \left(\frac{R}{I}\right)[x]$$

In particular, if I is a prime ideal of R, then this is a prime ideal of $R[x]$.

Proof. Define $\varphi : R[x] \to (R/I)[x]$ such that

$$\sum_{i=0}^{n} a_i x^i \mapsto \sum_{i=0}^{n} \bar{a_i} x^i$$

where $\bar{a_i} = a_i + I$. By the definition of multiplication and addition of cosets, φ is a ring homomorphism. Let $p(x) \in R[x]$ such that

$$p(x) = \sum_{i=0}^{n} a_i x^i$$
$$\bar{p(x)} = \sum_{i=0}^{n} \bar{a_i} x^i$$

We know

$$\begin{aligned} p(x) \in \ker \varphi &\iff \varphi(p(x)) = \bar{0} \\ &\iff \bar{p(x)} = \bar{0} \\ &\iff \bar{a_i} = \bar{0} \ \ \forall i \\ &\iff a_i \in I \ \ \forall i \end{aligned}$$

Thus $\ker \varphi = I[x] = (I)$. Hence the result follows from the first isomorphism theorem.
The second statement could be shown by letting I be a prime ideal in R. It follows that R/I is an integral

domain. Hence $(R/I)[x]$ is an integral domain. Since

$$\left(\frac{R}{I}\right)[x] \cong \frac{R[x]}{(I)}$$

we have that $R[x]/(I)$ is an integral domain. We conclude (I) is a prime ideal. □

Example 3.1.1. Consider $R = \mathbb{Z}$ and $I = n\mathbb{Z}$. Then

$$\frac{\mathbb{Z}[x]}{n\mathbb{Z}[x]} \cong \left(\frac{\mathbb{Z}}{n\mathbb{Z}}\right)[x]$$

If n is composite, then $(\mathbb{Z}/n\mathbb{Z})[x]$ is not an integral domain. Hence, $\mathbb{Z}[x]/n\mathbb{Z}[x]$ is not an integral domain. So $n\mathbb{Z}[x]$ is not prime. If n is a prime p, then

$$\left(\frac{\mathbb{Z}}{p\mathbb{Z}}\right)[x] \text{ is an integral domain.}$$

Hence, $\mathbb{Z}[x]/p\mathbb{Z}[x]$ is an integral domain, and so $p\mathbb{Z}[x]$ is a prime ideal of $\mathbb{Z}[x]$.

We now focus our attention to polynomials over a field. Let F be a field and define a norm on $F[x]$ by $N(p(x)) = deg(p(x)) \quad \forall p \in F[x]$.

Theorem 3.1.1. (Polynomails Over Fields are Euclidean Domains)
Let F be a field. Then $F[x]$ is a euclidean domain. Specifically, if $a, b \in F[x]$ such that $b(x) \neq 0$ then there exists unique $q, r \in F[x]$ such that

$$a(x) = q(x)b(x) + r(x)$$

where $r(x) = 0$ or $\deg(r) < \deg(b)$.

Proof. (Existence) We proceed by induction. If $a(x) = 0$, the $q(x) = 0$ and $r(x) = 0$ gives us what we want. If $a(x)$ is a nonzero constant, then we take $q(x) = 0$ and $r(x) = a(x)$ if $\deg(b) > 0$. We have $a(x) = 0 \cdot b(x) + a(x)$. If $b(x)$ is a nonzero constant then we write $b(x) = b$ and take $q(x) = a(x)/b$, $r(x) = 0$. Then $a(x) = q(x)b(x) + r(x)$.
Let $\deg(a(x)) = n$ and $\deg(b(x)) = \mathrm{m}$, and suppose the result holds for all polynomials of degree less than n. If $n < m$, we let $q(x) = 0$ and $a(x) = r(x)$. If $n \geq m$, let

$$a(x) = \sum_{i=0}^{n} a_i x^i = a_n x^n + ... + a_0$$
$$b(x) = \sum_{i=0}^{m} b_i x^i = b_m x^m + ... + b_0$$

where b_m and a_n are nonzero. Then

$$a'(x) = a(x) - \frac{a_n}{b_m} x^{n-m} \cdot b(x)$$

is a polynomial with $\deg(a') < \deg(a)$. By our inductive hypothesis

$$a'(x) = q'(x)b(x) + r(x)$$

for some $q'(x), r(x) \in F[x]$ where $r(x) = 0$ or $\deg(r) < \deg(b)$. Letting

$$q(x) = q'(x) + \frac{a_n}{b_m} x^{n-m}$$

we have

$$\begin{aligned} q(x)b(x) + r(x) &= \left(q'(x) + \frac{a_n}{b_m} x^{n-m}\right) b(x) + r(x) \\ &= q'(x)b(x) + \frac{a_n}{b_m} x^{n-m} b(x) + r(x) \\ &= a'(x) + \frac{a_n}{b_m} x^{n-m} b(x) \\ &= a(x) \end{aligned}$$

completing the induction.
(Uniqueness) Suppose $q_1(x)$ and $r_1(x)$ also satisfy the conditions of the theorem. Then

$$\begin{aligned} & q(x)b(x) + r(x) = q_1(x)b(x) + r_1(x) \\ \implies & b(x)(q(x) - q_1(x)) = r_1(x) - r(x) \end{aligned}$$

If $q(x) - q_1(x) \neq 0$, then

$$\deg[b(x)(q(x) - q_1(x))] = \deg(b(x)) + \deg(q(x) - q_1(x)) \geq m$$

However, $\deg(r_1(x) - r(x)) < m$. Hence,

$$\begin{aligned} & q(x) - q_1(x) = 0 \\ \implies & q(x) = q_1(x) \end{aligned}$$

It follows immediately that $r(x) = r_1(x)$. □

Corollary 3.1.1. If F is a field, then $F[x]$ is a principal ideal domain and a unique factorization domain.

This result follows from the fact that all euclidean domains are principal ideal domains, and all principal ideal domains are unique factorization domains.

Example 3.1.2. $\mathbb{Q}[x]$ is a principal ideal domain and a unique factorization domain.

Example 3.1.3. $\mathbb{Z}[x]$ is not a principal ideal domain. Examine the ideal $(2, x)$ in $\mathbb{Z}[x]$. Recall that

$$(2, x) = \{2p(x) + xq(x) : p(x), q(x) \in \mathbb{Z}[x]\}$$

These are polynomials with even constant coefficients.

$$(2, x) \neq \mathbb{Z}[x]$$

since $x + 1 \in \mathbb{Z}[x]$ but not in $(2, x)$.
Suppose there exists $d(x) \in \mathbb{Z}[x]$ such that $(d(x)) = (2, x)$. So $2 \in (d(x))$, hence there exists $f(x) \in \mathbb{Z}[x]$ such that

$$d(x)f(x) = 2 \text{ with } \deg(2) = 0$$

so $\deg(d(x)) = 0 = \deg(f(x))$ and

$$d(x) \in \{\pm 1, \pm 2\}$$

3.2 Irreducibility Criterion

Now that we can discuss the factorizations of polynomials over fields, we would look to discuss the irreducibility of these polynomials. Take a look at this example in the rationals.

Example 3.2.1. In $\mathbb{Q}[x]$, $7x$ is irreducible. We can only factor

$$7x = a(x)b(x)$$

where either $a(x)$ is a unit or $b(x)$ is a unit. Suppose $a(x)$ and $b(x)$ are nonunits. The units of $\mathbb{Q}[x]$ are exactly the units in $\mathbb{Q}$ so if $a(x)$ and $b(x)$ are nonunits they must have positive degree which implies

$$\deg(7x) > 2$$

a contradiction. However, $7x$ is reducible in $\mathbb{Z}[x]$: let $a(x) = 7$ and $b(x) = x$. We know that every intregral domain is contained in its field of fractions, for instance

$$\mathbb{Z} \subseteq \mathbb{Q} \implies \mathbb{Z}[x] \subseteq \mathbb{Q}[x]$$

We will take a look at theorems which relate irreducibility in $\mathbb{Q}[x]$ to irreducibility in $\mathbb{Z}[x]$, or more generally, factorization of polynomials in $R[x]$ with factorization in $F[x]$ (where F is the field of fractions of the integral domain R).

Lemma 3.2.1. (Gauss's Lemma Generalized)
Let R be a unique factorization domain with field of fractions F and let $p(x) \in R[x]$. If $p(x)$ is reducible in $F[x]$, then $p(x)$ is reducible in $R[x]$.

Lemma 3.2.2. (Gauss's Lemma Special)
Let $p(x) \in \mathbb{Z}[x]$. If $p(x)$ is reducible in $\mathbb{Q}[x]$, then $p(x)$ is reducible in $\mathbb{Z}[x]$.

Corollary 3.2.1. Let R be a unique factorization domain with field of fractions F. Let $p(x) \in R[x]$ and suppose all the coefficients of $p(x)$ are relatively prime. Then

$$p(x) \text{ is irreducible in } R[x] \iff p(x) \text{ is irreducible in } F[x]$$

We will work with polynomials with coefficients in a field F. Hence $F[x]$ is a euclidean domain, a principal ideal domain, and a unique factorization domain. The irreducible polynomials in $F[x]$ are all non-constants. In particular, all linear polynomials in $F[x]$ are irreducible.

Proposition 3.2.1. Let F be a field with $p(x) \in F[x]$. Then $p(x)$ has a linear factor if and only if $p(x)$ has a root in F.

Proof. If $p(x)$ has a linear factor, we may assume it is of the form $x - \alpha$ for some $\alpha \in F$. But then $p(x) = (x - \alpha)g(x)$ for some $g(x) \in F[x]$. It follows

$$p(\alpha) = (\alpha - \alpha)g(\alpha) = 0$$

Conversely, suppose α is a root of $p(x)$. Then by the division algorithm,

$$\exists q(x), r(x) \in F[x] \text{ such that } p(x) = q(x)(x - \alpha) + r(x)$$

where $r(x) = 0$ or $\deg(r) < 1$. Hence $r(x)$ is a constant polynomial. Since α is a root of $p(x)$,

$$p(\alpha) = (\alpha - \alpha)q(\alpha) + r(\alpha) = r(\alpha)$$

Since $r(x)$ is a constant, $r(x) = 0$. That is, $p(x)$ has a linear factor. □

Proposition 3.2.2. A polynomial of degree 2 or 3 is reducible if and only if it has a root in F.

We will need tools to find roots of polynomials.

Theorem 3.2.1. (Rational Root Theorem)
Let $p(x) = a_n x^n + ... + a_0 \in Z[x]$ where $\deg(p) = n$. If $n/s \in \mathbb{Q}$ with $\gcd(r, s) = 1$ and $p(r/s) = 0$, then $r \mid a_0$ and $s \mid a_n$. In particular, if $p(x)$ is monic ($a_n = 1$) in $\mathbb{Z}[x]$ and $p(d) \neq 0$ for all d such that $d \mid a_0$, then $p(x)$ has no roots in $\mathbb{Q}$.

Proof. Assume $p(r/s) = 0 = a_n(r/s)^n + ... + a_1(r/s) + a_0$. Multiplying through by s^n :

$$\begin{aligned} 0 &= a_n r^n + a_{n-1} r^{n-1} s + ... + a_1 r s^{n-1} + a_0 s^n \\ \implies a_0 s^n &= -a_n r^n - a_{n-1} r^{n-1} s - ... - a_1 r s^{n-1} \\ &= r(-a_n r^{n-1} - a_{n-1} r^{n-2} s - ... - a_1 s^{n-1}) \end{aligned}$$

Hence $r \mid a_0 s^n$, but $\gcd(r, s) = 1$ so $r \mid a_0$. Similarly,

$$\begin{aligned} a_n r^n &= -a_{n-1} r^{n-1} s - ... - a_1 r s^{n-1} - a_0 s^n \\ &= s(-a_{n-1} r^{n-1} - ... - a_1 r s^{n-2} - a_0 s^{n-1}) \end{aligned}$$

Hence $s \mid a_n r^n$, but $\gcd(r, s) = 1$ so $s \mid a_n$. □

Example 3.2.2. (i) $x^3 - 3x - 1$ is irreducible (over $\mathbb{Q}$) in $\mathbb{Q}[x]$. By the rational root theorem, a rational root would be ± 1, neither of which yield zero when plugged in.

(ii) $x^3 - 3x - 1$ in $(\mathbb{Z}/2\mathbb{Z})[x]$ is still irreducible.

(iii) $x^3 - 3x - 1$ in $(\mathbb{Z}/3\mathbb{Z})[x]$ is reducible since 1 is a root $\mod 3$.

Proposition 3.2.3. Let I be a prime ideal in the integral domain R. Let $p(x) \in R[x]$ be a non-constant polynomial with leading coefficient not in I. If $\bar{p(x)} \in (R/I)[x]$ cannot be factored in $(R/I)[x]$ into two polynomials of positive degree, then $p(x)$ cannot be factored similarly in $R[x]$.

Proof. Suppose $p(x)$ can be factored in $R[x]$. Then there exists $a(x), b(x) \in R[x]$ where $a(x)$ and $b(x)$ are non-constant polynomials in $R[x]$ such that $a(x)b(x) = p(x)$. Hence $\bar{a(x)} \cdot \bar{b(x)} = \bar{p(x)}$ and $\bar{a(x)}, \bar{b(x)}$ are nonconstant polynomials. Since the leading coefficient of $p(x)$ is not in I, the leading coefficients of $a(x)$ and $b(x)$ must also not be in I. Hence the leading coefficients are not $\bar{0}$ in R/I. □

Example 3.2.3. $p(x) = x^2 + x + 1 \in \mathbb{Z}[x]$. Examine $p(x)$ in $(\mathbb{Z}/2\mathbb{Z})[x]$ and notice $p(0) = 1$.

$$p(1) = 3 \equiv 1 \mod 2$$

Hence $p(x)$ is irreducible in $(\mathbb{Z}/2\mathbb{Z})[x] \implies p(x)$ is irreducible in $\mathbb{Z}[x] \implies p(x)$ is irreducible in $\mathbb{Q}[x]$.

Theorem 3.2.2. (Eisenstein's Criterion)
Let P be a prime ideal of an integral domain R. Let $f(x) = \sum_{i=0}^{n} a_i x^i \in R[x]$ be a polynomial with degree greater than 0. Suppose $a_0, ..., a_{n-1} \in P$, $a_n \notin P$, and $a_0 \in P^2$. Then $f(x)$ is irreducible in $R[x]$ ($P^2 = \{\sum_{i=1}^{n} x_i y_i : n \in \mathbb{Z}, x_i y_i \in P\}$).
In the integers: Let $p \in Z$ be prime and $f(x) = \sum_{i=0}^{n} a_i x^i \in \mathbb{Z}[x]$ with degree $n \geq 1$. Suppose $p \mid a_i \ \ \forall i \in \{0, 1, ..., n-1\}$, but $p \nmid a_n$ and $p^2 \nmid a_0$. Then $f(x)$ is irreducible* in $\mathbb{Z}[x]$, hence it is irreducible in $\mathbb{Q}[x]$.
$*f(x)$ cannot be factored into a product of two polynomials of positive degree.

Proof. Suppose that $f(x)$ can be written as a product of two polynomials of positive degree in $\mathbb{Z}[x]$. That is, $f(x) = a(x)b(x)$, $a(x), b(x) \in \mathbb{Z}[x]$ with $\deg(a) > 0$ and $\deg(b) > 0$. Consider $\bar{f(x)}, \bar{a(x)}$, and $\bar{b(x)}$ to be the reductions of $f(x), a(x)$, and $b(x)$ in $(\mathbb{Z}/p\mathbb{Z})[x]$ respectively. Since p divides all the coefficients of f except a_n, we have that

$$\bar{f(x)} = \bar{a_n} x^n \text{ since } \bar{f(x)} = \bar{a(x)}\bar{b(x)} = \bar{a(x)}\bar{b(x)}$$

We have $\bar{a(x)}\bar{b(x)} = \bar{a_n} x^n$. Since $\mathbb{Z}/p\mathbb{Z}$ is a field, $(\mathbb{Z}/p\mathbb{Z})[x]$ is a euclidean domain and hence a principal ideal domain, and further a unique factorization domain. Since x is an irreducible element of $(\mathbb{Z}/p\mathbb{Z})[x]$, we know that $\bar{a_n} x^n$ is the unique factorization up to associates of $\bar{f(x)}$ in $(\mathbb{Z}/p\mathbb{Z})[x]$. Thus $\bar{a(x)} = \bar{c} x^{n-r}$ and $\bar{b(x)} = \bar{b} x^r$ for some $b, c \in \mathbb{Z}$ where $\bar{b}\bar{c} = \bar{a_n}$ and $r < n$. Note $r \neq 0$ as this would imply that $\bar{b(x)}$ is constant and $r \neq n$ which implies $\bar{a(x)}$ is constant. The equation $\bar{a(x)} x^n = (\bar{c} x^{n-r})(\bar{b} x^r)$ implies that x

divides

$$\bar{c}x^{n-r} \text{ or } \bar{b}x^r$$

since x is also a prime.
$a(\bar{x})$ and $b(\bar{x})$ are the reductions of $a(x)$ and $b(x)$ in $(\mathbb{Z}/p\mathbb{Z})[x]$. This implies that the constant coefficients of both $a(\bar{x})$ and $b(\bar{x})$ are $\bar{0}$. That is, the constant coefficient of $a(x)$ and $b(x)$ must be divisible by p. Since $f(x) = a(x)b(x)$, the constant coefficient of $f(x)$ is divisible by p^2, a contradiiction. □

Proposition 3.2.4. *$F[x]/(p(x))$ is a field if and only if $p(x)$ is irreducible.*

Example 3.2.4. (i) $x^4 + 10x + 5 \in \mathbb{Z}[x]$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ since $5 \mid 10 = a_2$, $5 \mid 5 = a_0$, $5 \nmid 1 = a_4$, and $5^2 \nmid 5 = a_0$.

(ii) Let $a \in \mathbb{Z}$ such that $p \mid a$ but $p^2 \nmid a$ for some prime $p \in \mathbb{Z}$. Then $x^n - a$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ by Eisenstein's criterion for all $n \in \mathbb{N}$. In particular, $x^n - p$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$ for all $n \in \mathbb{N}$ and for all primes p. $x^3 - 7$ has no zeros in $\mathbb{Q}$, but has zeros outside of $\mathbb{Q}$:

$$(\sqrt[3]{7}, \sqrt[3]{7}w, \sqrt[3]{7}w^2) \text{ where } w = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$$

Example 3.2.5. $f(x) = x^4 + 1$ is irreducible over $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$, but Eisenstein's criterion does not apply... or does it? Let's examine:

$$g(x) = f(x+1) = (x+1)^4 + 1 = x^4 + 4x^3 + 6x^2 + 4x + 2$$

We see that $g(x)$ is irreducible by Eisenstein's criterion with $p = 2$. Suppose $f(x) = a(x)b(x)$ where $a(x), b(x) \in \mathbb{Z}[x]$ with positive degrees. Then

$$f(x) = a(x)b(x) \iff f(x+1) = a(x+1)b(x+1)$$

Example 3.2.6. (Important Example) Cyclotomic Polynomial. Let p be a prime in $\mathbb{Z}$. Consider

$$\phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \ldots + x + 1$$

Then

$$\begin{aligned}\phi_p(x+1) = \frac{(x+1)^p - 1}{x+1-1} &= \frac{\sum_{k=0}^{p} \binom{p}{k} x^k 1^{p-k} - 1}{x} \\ &= \frac{\sum_{k=1}^{p} \binom{p}{k} x^k}{x} \\ &= \sum_{k=1}^{p} \binom{p}{k} x^{k-1}\end{aligned}$$

Then $a_0 = k$, $a_n = 1$, and $p \mid a_k$ for all $0 \leq k < n$, $p \nmid a_n$, and $p^2 \mid a_0$. By Eisenstein's criterion, $\phi_p(x+1)$ is irreducible in $\mathbb{Z}[x]$ and $\mathbb{Q}[x]$.

3.3 Vector Spaces

We now briefly take some time to discuss vector spaces. We will introduce some basic definitions and properties without going into too much detail. We are setting up the necessary background for field extensions.

Definition 3.3.1. (Vector Space)
Let F be a field. A vector space over F is F with a set V together with

(i) A binary operation $+$ on V such that $< V, + >$ is an abelian group.

(ii) An action of F on V (a mapping $f \times V$ to V), denoted kv for every $k \in F$ and $v \in V$, which satisfies

(a) $(j+k)v = jv + kv$ for all $j, k \in F, v \in V$
(b) $(jk)v = j(kv)$ for all $j, k \in F, v \in V$
(c) $j(v+w) = jv + jw$ for all $j \in F, v, w \in V$
(d) $1v = v$ for all $v \in V$

Definition 3.3.2. (Finite Linear Combination)
Given $v_1, ..., v_n \in V$ a vector space and $\alpha_1, ..., \alpha_n \in F$ a field, a sum of the form

$$\alpha_1 v_1 + ... + \alpha_n v_n$$

is a finite linear combination of $v_1, ..., v_n$.

Definition 3.3.3. (Span of a Set)
Given $S \subseteq V$, the span of S, denoted span(S), is the set of all finite linear combinations (lc) of elements in S:

$$\text{span}(S) = \left\{ \sum_{i=1}^{n} a_i s_i : n \in \mathbb{N}, \alpha_i \in F, s_i \in S \right\}$$

Definition 3.3.4. (Linear Independence)
A subset $S \subseteq V$ is called a set of linearly independent (li) vectors if the equation

$$\alpha_1 v_1 + ... + \alpha_n v_n = 0 \text{ for all } \alpha_i \in F, v_i \in S$$

implies that $\alpha_1 = ... = \alpha_n = 0$.

Definition 3.3.5. (Basis)
A basis of a vector space V is an ordered set of linearly independent vectors B which span V. That is,

$$\text{span}(B) = V$$

Proposition 3.3.1. Assume that $A = \{v_1, ..., v_n\} \subseteq V$ is spans V, but every $S \subset A$ does not span V. Then A is a basis of V.

Proof. We know span$(A) = V$. It remains to check that $v_1, ..., v_n$ are linearly independent vectors. Suppose for the sake of contradiction

$$\alpha_1 v_1 + ... + \alpha_n v_n = 0$$

for $\alpha_i \in F$ and not all α_i are zero. Assume without loss of generality that $\alpha_1 \neq 0$. Then

$$v_1 = \frac{1}{\alpha_1}(-\alpha_2 v_2 - ... - \alpha_n v_n) \qquad (*)$$

It follows that

$$\text{span}(v_2, ..., v_n) = V$$

since any linear combination of $v_1, ..., v_n$ can be written as a linear combination of $v_2, ..., v_n$ using $(*)$. Hence we have arrived at a contradiction. □

Example 3.3.1. Let F be a field and $p(x) \in F[x]$ be an irreducible polynomial. From our homework, we know that $F[x]/(p(x))$ is a field. We were told that $F[x]/(p(x))$ is a vector space over F. You check the details (scalar multiplication $k(f(x)+g(x)) = kf(x)+kg(x)$). We also found out in our homework that

$$\text{span}(\{\bar{1}, \bar{x}, \bar{x^2}, ..., \bar{x^{n-1}}\}) = \frac{F[x]}{(p(x))}$$

To prove the claim $\bar{1}, ..., \bar{x^{n-1}}$ forms a basis, we only need to show the linear independence of $\bar{1}, ..., \bar{x^{n-1}}$. To that end, consider

$$\alpha_0\bar{1} + ... + \alpha_{n-1}\bar{x^{n-1}} = \bar{0}$$

with not all $\alpha_i = 0$. Then

$$\begin{aligned} &\alpha_0 + \alpha_1 x + ... + \alpha_{n-1}x^{n-1} + (p(x)) = 0 + (p(x)) \\ \implies &\alpha_0 + \alpha_1 x + ... + \alpha_{n-1}x^{n-1} \in (p(x)) \\ \implies &p(x) \mid (\alpha_0 + \alpha_1 x + ... + \alpha_{n-1}x^{n-1}) \end{aligned}$$

However, $\deg(p(x)) = n$ but this is impossible uless $a_0 + a_1x + ... + a_{n-1}x^{n-1}$ is the zero polynomial. Hence $\alpha_0 = ... = \alpha_{n-1} = 0$ so $\bar{1}, ..., \bar{x^{n-1}}$ are linearly independent vectors.

Corollary 3.3.1. Assume the set $A \subseteq V$ spans V. Then A contains a basis of V.

Proof. If A is a basis, we are done. Suppose A is not a basis. Then there exists $A_1 \subset A$ such that $\text{span}(A_1) = V$. If A_1 is a basis, we are done. Otherwise, there exists $A_2 \subset A_1$ such that $\text{span}(A_2) = V$. Since our containments are proper and finite, this process must terminate. □

Theorem 3.3.1. (Replacement Theorem)
Suppose $A = \{a_1, ..., a_n\} \subseteq V$ is a basis for V containing n elements and $B = \{b_1, ..., b_m\} \subseteq V$ is a set of linearly independent vectors in V. Then there is an ordering of $a_1, ..., a_n$ such that for each $k \in \{1, ..., m\}$ the set $\{b_1, ..., b_k, a_{k+1}, ..., a_n\}$ is a basis of V. In particular, $m \leq n$.

Proof. We will use induction on k.
If $k = 0$, then there is nothing to do since A is assumed to be a basis of V. Suppose that $B_k = \{b_1, ..., b_k, a_{k+1}, ..., a_n\}$ is a basis for V. Then $\text{span}(B_k) = V$, so

$$b_{k+1} = \beta_1 b_1 + ... + \beta_k b_k + \alpha_{k+1}a_{k+1} + ... + \alpha_n a_n$$

for some $\beta_i, \alpha_i \in F$. Note that not all α_j equal zero, otherwise b_{k+1} is linearly independent of $b_1, ..., b_k$, contradicting the linear independence of B. Without loss of generality, assume $\alpha_{k+1} \neq 0$. Then

$$a_{k+1} = \frac{1}{\alpha_{k+1}}\left(b_{k+1} - (\beta_1 b_1 + ... + \beta_k b_k + \alpha_{k+1}a_{k+1} + ... + \alpha_n a_n)\right)$$

Hence,

$$\text{span}(b_1, ..., b_k, b_{k+1}, a_{k+2}, ..., a_n) = \text{span}(b_1, ..., b_k, a_{k+1}, a_{k+2}, ..., a_n) = V$$

It remains to show that

$$B_{k+1} = \{b_1, ..., b_k, b_{k+1}, a_{k+2}, ..., a_n\}$$

is a linear independent set. If

$$\beta_1' b_1 + ... + \beta_{k+1}' b_{k+1} + \alpha_{k+2}' a_{k+2} + ... + \alpha_n' a_n = 0$$

then

$$\beta_1' b_1 + ... + \beta_{k+1}'(\beta_1 b_1 + ... + \beta_k b_k + \alpha_{k+1}a_{k+1} + ... + \alpha_n a_n) + \alpha_{k+2}' a_{k+2} + ... + \alpha_n' a_n = 0$$

Noting that the right hand side of this equation is in B_k, we know that all (simplified) coefficients must be 0 since B_k is assumed to be linearly independent. In particular, the coefficient of a_{k+1} is $\beta_{k+1}'\alpha_{k+1} = 0$ so $\beta_{k+1}' = 0$ or $\alpha_{k+1} = 0$ since fields are integral domains. Recall $\alpha_{k+1} \neq 0$, hence $\beta_{k+1}' = 0$. But then we have

$$\beta_1' b_1 + ... + \beta_k' b_k + \alpha_{k+2}' a_{k+2} + ... + \alpha_n' a_n = 0$$

Thus all $\beta_i', \alpha_j' \in F$ must be zero since $b_1, ..., b_k, a_{k+1}, ..., a_n$ are linearly independent by the inductive hypothesis. □

Corollary 3.3.2. (i) Suppose V is a finite basis with n elements. Any set of linearly independent vectors has at most n elements.

(ii) If V has some finite basis, then any two bases of V have the same number of elements.

Definition 3.3.6. (Dimension of a Vector Space)
If V is a vector space over F with a finite basis, then the cardinality of any basis is called the dimension of V, denoted $\dim V$.

Chapter 4

Field Theory

4.1 Extension Fields

When studying irreducibility criteria for polynomials, one question that arises is whether we can find a root in the field we are working in. For example, the polynomial $x^2 - 2$ has no roots in $\mathbb{Q}$. Could we find a way to enlarge $\mathbb{Q}$ so that $x^2 - 2$ has a root while also preserving the field structure? This leads us to the concept of a field extension. In this section, we introduce field extensions and some of their properties.

We begin by first by recalling that a field F is a commutative ring with identity such that every nonzero element has an inverse. Equivalently, the set $F^\times = F \setminus \{0\}$ is an abelian group under multiplication. Associated with any field F is its characteristic.

Definition 4.1.1. (Characteristic of a Field)
Recall that the characteristic of a field F, denoted $ch(F)$, is the additive order of 1 in F. If the order of 1 is infinite, we say $ch(F) = 0$.

Proposition 4.1.1. $ch(F) = 0$ or p where p is prime. If $ch(F) = p$, then

$$\forall \alpha \in F \quad p\alpha = 0$$

Proof. If $ch(F) = 0$, we are done. Suppose $ch(F) \neq 0$. Suppose $ch(F) = n$ where n is composite. Then $n = a \cdot b$ where $1 < a < n$ and $1 < b < n$. Then

$$\begin{aligned}(a \cdot 1_F)(b \cdot 1_F) &= \underbrace{(1_F + ... + 1_F)}_{a\text{-times}}\underbrace{(1_F + ... + 1_F)}_{b\text{-times}} \\ &= \underbrace{1_F + ... + 1_F + 1_F + ... + 1_F}_{ab\text{-times}} \\ &= ab \cdot 1_F \\ &= 0\end{aligned}$$

Since fields are integral domains, this gives $a \cdot 1_F = 0$ or $b \cdot 1_F = 0$. Either of which contradicts the minimality of $ch(F)$. □

Example 4.1.1. (i) $ch(\mathbb{Q}) = 0 = ch(\mathbb{R})$.

(ii) Denote $\mathbb{Z}/p\mathbb{Z}$ by $\mathbb{F}_p$. Then $ch(\mathbb{F}_p) = p$.
In the homework, we explored the homomorphism

$$\begin{aligned}\phi : \mathbb{Z} &\to F \\ \phi(n) &= n \cdot 1_F\end{aligned}$$

defined by

$$n \cdot 1 = 0 \text{ when } n = 0$$
$$n \cdot 1 = \underbrace{1 + ... + 1}_{n\text{-times}} \text{ when } n > 0$$
$$n \cdot 1 = \underbrace{-1 - ... - 1}_{n\text{-times}} \text{ when } n < 0$$

$\ker \phi = n\mathbb{Z}$ where $n = ch(F)$. The first isomorphism theorem says

$$\frac{\mathbb{Z}}{\ker \phi} \cong \phi(\mathbb{Z})$$

That is,

$$\frac{\mathbb{Z}}{ch(F)} \cong \phi(\mathbb{Z}) \subseteq F$$

Then when $ch(F) = 0$ we have

$$Z \cong \frac{\mathbb{Z}}{\{0\}} \cong \phi(\mathbb{Z})$$

so a field of characteristic 0 contains an isomorphic copy of $\mathbb{Z}$ and thus also contains an isomorphic copy of $\mathbb{Q}$. In the case that $ch(F) = p$ a prime, we have

$$\frac{\mathbb{Z}}{p\mathbb{Z}} \cong \phi(\mathbb{Z})$$

Hence any field with characteristic p a prime contains an isomorphic copy of $\mathbb{Z}/p\mathbb{Z}$.

Definition 4.1.2. (Prime Subfield)
The prime subfield of a field F is the subfield of F generated by the multiplicative identity of F, 1_F. It is isomorphic to $\mathbb{Q}$ (if $ch(F) = 0$) or $\mathbb{F}_p$ (if $ch(F) = p$ a prime).

Definition 4.1.3. (Extension Field)
If K is a field containing the subfield F, then K is said to be an extension field of F (or an extension of F) denoted K/F (read K over F, NOT $K \mod F$), or by the diagram:

$$\begin{array}{c} K \\ | \\ F \end{array}$$

In particular, every field is an extension of its prime subfield.

If K/F is any extension of fields, then K is a vector space over F. Note that $\langle K, + \rangle$ is an abelian group and since $F \subseteq K$ the scalar multiplication is defined by the multiplication in K. In particular, every field F can be considered a vector space over its prime subfield.

Definition 4.1.4. (Degree of a Field Extension)
The degree (or relative degree or index) of a field extension K/F, denoted $[K : F]$, is the dimension of K as a vector space over F. The extension is said to be finite if $[K : F]$ is finite, otherwise we say the extension is infinite.

Our first big result about field extensions: If we have an irreducible polynomial $p(x) \in F[x]$ we can find a field K containing F where $p(x)$ has a zero.

Recall . For $\phi : F \to F'$ a homomorphism of fields we have ϕ is either identically zero or ϕ is injective. So

$$\phi(F) = 0 \text{ or } \phi(F) \cong F$$

Proposition 4.1.2. Let F be a field and let $p(x) \in F[x]$ be an irreducible polynomial. Then there exists a field K containing an isomorphic copy of F in which $p(x)$ has a root.

Proof. Consider the quotient

$$K = \frac{F[x]}{(p(x))}$$

We have discovered that $(p(x))$ is maximal and K is a field. Letting $\phi = \pi|_F : F \to K$ (where π is the natural projection from $F[x]$ to $F[x]/(p(x))$ restricted to the constant polynomials in $F[x]$, where $\phi(f) = \pi(f) = f + (p(x)) \quad \forall f \in F$) gives a homomorphism of fields. Note that this mapping is not identically zero since $\phi(1_F) = 1_F + (p(x)) \neq 0 + (p(x))$ since K is a field and $1_F \neq 0$. Hence

$$\phi(F) \cong F$$

Therefore $\phi(F)$ is our isomorphic copy of F in K. We now consider F a subfield of K. Note then that there is an isomorphic copy of $F[x]$ contained in $K[x]$. Let $\bar{x} = \pi(x) = x + (p(x)) \in K$. Then $p(\bar{x}) = \bar{p(x)}$ since π is a homomorphism. Then

$$p(\bar{x}) = p(x) + (p(x)) = 0 + (p(x))$$

so $\bar{x} \in K$ and $\bar{x}$ is the root of $p(x)$. Hence, K is an extension of F in which $p(x)$ has a root. □

Theorem 4.1.1. Let $p(x) \in F[x]$ be an irreducible polynomial of degree n. Let $\theta = x \mod p(x)$ in K. Then the elements

$$1, \theta, \theta^2, ..., \theta^{n-1}$$

form a basis for K as a vector space over F so the degree of the extension K/F is n, i.e. $[K : F] = n$. So $K = \left\{a_0 + a_1\theta + a_2\theta^2 + ... + a_{n-1}\theta^{n-1} : a_0, ..., a_{n-1} \in F\right\}$ consists of all polynomials of degree less than n.

Example 4.1.2. Let $F = \mathbb{R}$ and $p(x) = x^2 + 1$. $p(x)$ is irreducible in $\mathbb{R}$. The field

$$K = \frac{\mathbb{R}[x]}{(x^2 + 1)}$$

is an extension of degree 2. Elements of K are of the form $a+b\theta$ such that $a, b \in \mathbb{R}$ and $\theta \equiv x \mod (x^2+1)$. Define

$$\begin{aligned} \varphi : \frac{\mathbb{R}}{(x^2 + 1)} &\to \mathbb{C} \\ a + b\theta &\mapsto a + bi \end{aligned}$$

φ is a homomorphism which is clearly surjective and $\ker \varphi = \{0\}$.

4.2 Simple Extensions

We ended the previous section by demonstrating that adjoining a root of an irreducible polynomial to a field F produces an extension whose elements are polynomials in that root with a bounded degree. We focus on studying the extensions generated by a single element. First, we define what it means to generate a field.

Definition 4.2.1. (Generated Field)
Let K/F be an extension of fields and let $\alpha, \beta, \gamma, ... \in K$. Then the smallest subfield of K containing both F and $\alpha, \beta, \gamma, ...$ is denoted $F(\alpha, \beta, \gamma, ...)$ is called the field generated by $\alpha, \beta, \gamma, ...$ over F.

Definition 4.2.2. (Simple Extension, Primitive Element)
If K is generated by a single element α over F, $K = F(\alpha)$, then K is said to be a simple extension of F and α is called a primitive element of the extension.

By adjoining a root α of an irreducible polynomial $p(x)$ to F, we get a simple extension $F(\alpha)$. Furthermore, this extension is isomorphic to the quotient $F[x]/(p(x))$.

Theorem 4.2.1. Let F be a field and let $p(x) \in F[x]$ be irreducible over F. Suppose K is an extension field of F containing a root α of $p(x)$. Then

$$F(\alpha) \cong \frac{F[x]}{(p(x))}$$

Proof. Consider $\varphi : F[x] \to F[\alpha]$ where $a(x) \mapsto a(\alpha)$. Here, $F[\alpha]$ are polynomials in α with coefficients in F. Note that, by closure, $F[\alpha] \subseteq F(\alpha)$ and since $F \subseteq K$ and $\alpha \in K$, we have $F(\alpha) \subseteq K$. Since addition and multiplication are defined exactly the same in $F[x]$ and $F[\alpha]$, φ is a ring homomorphism. Since α is a root of $p(x)$,

$$\begin{aligned} &\varphi(p(x)) = p(\alpha) = 0 \\ \implies\ &p(x) \in \ker \varphi \\ \implies\ &(p(x)) \subseteq \ker \varphi \end{aligned}$$

We use this to induce a well-defined homomorphism from $F[x]/(p(x))$ to $F(\alpha)$. Define $\mu : F[x]/(p(x)) \to F(\alpha)$ by $a(x) + (p(x)) \mapsto \varphi(\alpha)$. Note

$$a(x) + (p(x)) = b(x) + (p(x)) \implies a(x) = b(x) + q(x)p(x)$$

for some $q(x) \in F[x]$. So

$$\begin{aligned} \mu(a(x) + (p(x))) &= \mu(b(x) + q(x)\,(p(x) + (p(x)))) \\ &= \varphi(b(x) + q(x)(x)) \\ &= b(\alpha) + q(\alpha)p(\alpha) \\ &= b(\alpha) \\ &= \varphi(b(x)) \\ &= \mu(b(x) + (p(x))) \end{aligned}$$

Since the domain of μ is a field,

$$\ker \mu = \{0\} \ \text{ or } \ \ker \mu = \frac{F[x]}{(p(x))}$$

However, $\mu(1 + (p(x))) = 1 \neq 0$ so μ is not identically zero. It follows that $\ker \mu = \{0\}$ and μ is injective. The first isomorphism theorem then gives

$$\frac{F[x]}{(p(x))} \cong \mu\left(\frac{F[x]}{(p(x))}\right) \subseteq F(\alpha)$$

Noting that $\mu(x + (p(x))) = \alpha$ and $\mu(m + (p(x))) = m$ for all $m \in F$. Hence $\mu\,(F[x]/(p(x)))$ contains both F and α. Thus by definition $F(\alpha) \subseteq \mu\,(F[x]/(p(x)))$. Therefore, μ is onto and further it is an isomorphism. □

Corollary 4.2.1. Suppose in the previous theorem $\deg(p(x)) = n$. Then

$$F(\alpha) = \left\{a_0 + a_1\alpha + a_2\alpha^2 + ... + a_{n-1}\alpha^{n-1} : a_0, ..., a_{n-1} \in F\right\}$$

Example 4.2.1. Consider $\alpha = \sqrt[3]{2} \in \mathbb{R}$. Then α is a root of $x^3 - 2$ which is irreducible over $\mathbb{Q}$ by Eisenstein's criterion. So

$$\mathbb{Q}(\sqrt[3]{2}) = \left\{a + b\sqrt[3]{2} + c\left(\sqrt[3]{2}\right)^2 : a, b, c, \in \mathbb{Q}\right\}$$

$$\frac{\mathbb{Q}[x]}{(x^3 - 2)} \cong \mathbb{Q}(\sqrt[3]{2})$$

Furthermore, $x^3 - 2$ has roots $\sqrt[3]{2}\omega$ and $\sqrt[3]{2}\omega^2$ where $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$.

$$\mathbb{Q}\left(\sqrt[3]{2}\omega\right) = \left\{a + b\left(\sqrt[3]{2}\omega\right) + c\left(\sqrt[3]{2}\omega\right)^2 : a, b, c \in \mathbb{Q}\right\} \cong \frac{\mathbb{Q}[x]}{(x^3 - 2)}$$

As demonstrated in the example above, the roots of an irreducible polynomial are algebraically indistinguishable in the sense that they generate isomorphic extensions. The fields were distinguished not by their algebraic properties, but by whether their elements were real or complex.

4.3 Algebraic Extensions

Recall that i is a root of the polynomial x^2+1 in $\mathbb{Q}[x]$, so we say that i is algebraic. π, however, is not a root of any nonzer polynomial in $\mathbb{Q}[x]$ and so we say that π is not algebraic over $\mathbb{Q}$. π is a root of the polynomial $x-\pi$, so π is algebraic over $\mathbb{R}$.

Definition 4.3.1. (Algebraic, Transcendental)
The element $\alpha \in K$ is algebraic over F if α is a root of any nonzero polynomial in $F[x]$. If α is not algebraic over F, then α is said to be transcendental over F. The extension K/F is algebraic if every element of K is algebraic over F.

Proposition 4.3.1. Let α be algebraic over F. Then there is a unique, monic, irreducible polynomial $m_{\alpha,F} \in F[x]$ which has α as a root. A polynomial $f(x) \in F[x]$ has α as a root if and only if $m_{\alpha,F}(x)$ divides $f(x)$ in $F[x]$.

Proof. Let $g(x) \in F[x]$ be a polynomial with minimal degree which has α as a root. Multiplying $g(x)$ by a constant will give a monic polynomial, so without loss of generality assume $g(x)$ is monic. Suppose $g(x)$ is irreducible in $F[x]$, say $g(x) = a(x)b(x)$ where $0 < \deg a(x), \deg b(x) < \deg g(x)$. Then

$$g(\alpha) = a(\alpha)b(\alpha) = 0$$

Since K is a field and hence an integral domain, then $a(\alpha) = 0$ or $b(\alpha) = 0$, contradicting the minimality of $\deg g(x)$. It follows that $g(x)$ is a monic, irreducible polynomial in $F[x]$ with α as a root. Suppose now that $f(x) \in F[x]$ has α as a root. By the division algorithm

$$\exists q(x), r(x) \in F[x] \text{ such that } f(x) = q(x)g(x) + r(x)$$

where $r(x) = 0$ or $\deg r(x) < \deg g(x)$. Then

$$\begin{aligned} 0 = f(\alpha) &= q(\alpha)g(\alpha) + r(\alpha) \\ &= 0 + r(\alpha) \\ &= r(\alpha) \end{aligned}$$

By the minimality of $\deg g(x)$, $r(x) = 0$ since α is a root of $r(x)$. hence $g(x)$ divides any polynomial with α as a root in $F[x]$. If $g(x) \mid h(x)$ where $h(x) \in F[x]$, then α is a root of $h(x)$.
Since $g(x)$ divides any polynomial with α as a root, $g(x)$ divides monic polynomials with α as a root. That is, for any monic polynomial $p(x) \in F[x]$ with α as a root which has minimal degree

$$c \cdot g(x) = p(x)$$

so $g(x)$ is unique, and we let $g(x) = m_{\alpha,F}(x)$ from the statement. □

Definition 4.3.2. (Minimal Polynomial)
The polynomial $m_{\alpha,F}(x)$ (or just $m_\alpha(x)$ if F is understood) in the previous proposition is called the minimal polynomial of α over F. The degree of $m_\alpha(x)$ is called the degree of α.

Proposition 4.3.2. Let α be algebraic over F and let $F(\alpha)$ be the field generated by α over F. Then

$$F(\alpha) \cong \frac{F[x]}{(m_{\alpha,F}(x))}$$

and

$$[F(\alpha) : F] = \deg(m_{\alpha,F}(x)) = \deg(\alpha)$$

Example 4.3.1.
(i) Let $\alpha = \sqrt{2}$ and $F = \mathbb{Q}$. Then the minimal polynomial of α over $\mathbb{Q}$ is $m_\alpha(x) = x^2 - 2$. Hence $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = \deg(m_\alpha(x)) = 2$.

(ii) Let $\alpha = \sqrt[n]{2}$ and $F = \mathbb{Q}$. Then the minimal polynomial of α over $\mathbb{Q}$ is $m_\alpha(x) = x^n - 2$. Hence $[\mathbb{Q}(\sqrt[n]{2}) : \mathbb{Q}] = \deg(m_\alpha(x)) = n$.

Example 4.3.2. Let $p(x) = x^3 - 3x - 1$. This is a cubic so it has at least one real root. By the rational root theorem,

$$\left.\begin{array}{l} p(1) = -3 \\ p(-1) = 1 \end{array}\right\} \implies p(x) \text{ is irreducible over } \mathbb{Q}.$$

Let α denote the real root of $p(x)$. Then

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3.$$

We can now deduce the degree of a simple extension through the degree of the minimal polynomial. The next result relates the degree of a simple extension to whether the element generating the extension is algebraic.

Proposition 4.3.3. The element α is algebraic over F if and only if the simple extension $F(\alpha)$ is finite. More precisely, if α is an elment of an extension of degree n over F, then α satisfies a polynomial of degree at most n over F and if α satisfies a polynomial of degree n over F, then the degree of $F(\alpha)$ over F is at most n.

Proof. If α is algebraic over F, then α satisfies a polynomila of positive degree, say n, over F. Thus

$$[F(\alpha) : F] = \deg(m_{\alpha,F}(x)) \leq n$$

since $m_{\alpha,F}(x)$ divides any polynomial satisfied by α.
Conversely, suppose α is an element of an extension of degree n over F. Then the $n+1$ elements $1, \alpha, \alpha^2, ..., \alpha^n$ of $F(\alpha)$ are linearly independent over F. That is, we have

$$b_0 + b_1\alpha + b_2\alpha^2 + ... + b_n\alpha^n = 0$$

for some $b_i \in F$ with not all zero. Hence α is a root of

$$p(x) = b_0 + b_1x + b_2x^2 + ... + b_nx^n \in F[x]$$

where $p(x) = 0$ and $\deg(p(x)) \leq n$. Hence α is algebraic over F. □

Corollary 4.3.1. If the extension K/F is finite, then it is algebraic.

Proof. If $\alpha \in K$, then the subfield $F(\alpha)$ is a subspace of the vector space K over F. Hence, $[F(\alpha) : F] \leq [K : F]$. Since $[K : F] \in \mathbb{N}$, $F(\alpha)$ is a finite extension. Thus α is algebraic. □

Theorem 4.3.1. (Tower Law)
Let $F \subseteq K \subseteq L$ be fields. Then

$$[L : F] = [L : K][K : F]$$

Proof. Suppose $[L : K] = m$ and $[K : F] = n$ for $m, n \in \mathbb{N}$. Let $\alpha_1, ..., \alpha_m$ be a basis for L over K and let $\beta_1, ..., \beta_n$ be a basis for K over F. Then every element of L can be written as

$$a_1\alpha_1 + ... + a_m\alpha_m \text{ for } a_i \in K$$

Since a_i are elements of K we have

$$a_i = b_{i1}\beta_1 + b_{i2}\beta_2 + ... + b_{in}\beta_n \text{ for } b_{ij} \in F$$

Our elements can be written

$$\left(\sum_{j=1}^{n} b_{1j}\beta_1\right)\alpha_1 + \left(\sum_{j=1}^{n} b_{2j}\beta_2\right)\alpha_2 + ... + \left(\sum_{j=1}^{n} b_{mj}\beta_m\right)\alpha_m = \sum_{i=1}^{m}\sum_{j=1}^{n} b_{ij}\beta_j\alpha_i$$

where there are $m \cdot n$ elements $\beta_j\alpha_i$ and b_{ij} are in F. Hence, $\beta_j\alpha_i$ span L as a vector space over F. Suppose now that

$$\sum_{i=1}^{m}\sum_{j=1}^{n} b_{ij}\beta_j\alpha_i = 0 \text{ for } b_{ij} \in F$$

Then since the β_j form a basis for K over F, the equation above can be written as

$$a_1\alpha_1 + ... + a_m\alpha_m = 0$$

where $a_i \in K$ since

$$\left(\sum_{j=1}^{n} b_{ij}\beta_j\right) \in K$$

Since the $\alpha_i's$ form a basis for L over K, it follows that $a_i = 0$ for all i. Thus

$$b_{i1}\beta_1 + ... + b_{in}\beta_n = 0 \text{ for } 1 \leq i \leq m$$

Since the $\beta_j's$ form a basis of K over F, this implies that $b_{ij} = 0$ for all i, j. Hence $\alpha_i\beta_j$ are all linearly independent over F and hence form a basis for L over F. That is,

$$[L : F] = [L : K][K : F].$$

The cases where one of the extensions is infinite is in the book. □

Corollary 4.3.2. Suppose L/F is a finite extension and let K be a subfield of L containing F, $F \subseteq K \subseteq L$. Then

$$[K : F] \mid [L : F]$$

Example 4.3.3. Let $\alpha = 1 + \sqrt[3]{2} + \sqrt[3]{4}$. What is the degree of the extension $\mathbb{Q}(\alpha)$?

$$\alpha = 1 + \sqrt[3]{2} + \left(\sqrt[2]{2}\right)^2 \in \mathbb{Q}(\sqrt[3]{2})$$

Since $\alpha \in \mathbb{Q}(\sqrt[3]{2})$, $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\sqrt[3]{2})$. Thus

$$3 = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(\alpha)][\mathbb{Q}(\alpha) : \mathbb{Q}]$$

Consider $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(\alpha)] = 3$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 1$. Then

$$[\mathbb{Q}(\alpha) : \mathbb{Q}] = 1 \implies \mathbb{Q}(\alpha) = \mathbb{Q}$$

However, $\alpha \notin \mathbb{Q}$. Then it must be the case that $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}(\alpha)] = 1$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Hence

$$\mathbb{Q}(\sqrt[3]{2}) = \mathbb{Q}(\alpha)$$

Example 4.3.4. Let α be the real root of $x^3 - 3x - 1$. We will show $\sqrt{2} \notin \mathbb{Q}(\alpha)$. That is, we will show $\sqrt{2} \neq a + b\alpha + c\alpha^2$ for any $a, b, c \in \mathbb{Q}$. For the sake of contradiction, suppose $\sqrt{2} \in \mathbb{Q}(\alpha)$. Then

$$\mathbb{Q}(\sqrt{2}) \subseteq \mathbb{Q}(\alpha) \implies [\mathbb{Q}(\alpha) : \mathbb{Q}] = [\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{2})][\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = [\mathbb{Q}(\alpha) : \mathbb{Q}(\sqrt{2})] \cdot 2.$$

Hilariously, $2 \mid 3$, a contradiction.

Example 4.3.5. $\sqrt[6]{2}$ is the positive sixth root of 2. Since

$$m_{\sqrt[6]{2},\mathbb{Q}}(x) = x^6 - 2$$

we have $[\mathbb{Q}(\sqrt[6]{2}) : \mathbb{Q}] = 6$. Consider $\left(\sqrt[6]{2}\right)^3 = \sqrt{2} \in \mathbb{Q}\left(\sqrt[6]{2}\right)$. Thus

$$\begin{aligned} &\mathbb{Q}\left(\sqrt{2}\right) \subseteq \mathbb{Q}\left(\sqrt[6]{2}\right) \\ \implies &6 = \left[\mathbb{Q}\left(\sqrt[6]{2}\right) : \mathbb{Q}\right] = \left[\mathbb{Q}\left(\sqrt[6]{2}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right]\left[\mathbb{Q}\left(\sqrt{2}\right) : \mathbb{Q}\right] = 2 \\ \implies &\left[\mathbb{Q}\left(\sqrt[6]{2}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right] = 3 \end{aligned}$$

Example 4.3.6. Let $\alpha = \sqrt[5]{2} + 1$. Find the degree of $\mathbb{Q}(\alpha)$.
Notice

$$\begin{aligned} \alpha \in \mathbb{Q}(\sqrt[5]{2}) = \left\{a + b\sqrt[5]{2} + c\sqrt[5]{4} + d\sqrt[5]{8} + e\sqrt[5]{16} : a, b, c, d, e \in \mathbb{Q}\right\} \\ \implies \mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\sqrt[5]{2}) \end{aligned}$$

Also,

$$\left[\mathbb{Q}(\sqrt[5]{2}) : \mathbb{Q}\right] = 5$$

since $m_{\alpha,\mathbb{Q}}(x) = x^5 - 2$. Since $\mathbb{Q}(\alpha)$ is a field, $\alpha - 1 \in \mathbb{Q}(\alpha)$. That is,

$$\sqrt[5]{2} \in \mathbb{Q}(\alpha) \implies \mathbb{Q}(\sqrt[5]{2}) \subseteq \mathbb{Q}(\alpha)$$

Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt[5]{2})$, so the degree of $\mathbb{Q}(\alpha) = 5$.

Lemma 4.3.1. $F(\alpha, \beta) = (F(\alpha))(\beta)$. I.e., the field generated by α and β over F is the field generated by β over $F(\alpha)$.

Proof. Note that $F \subseteq F(\alpha, \beta), \alpha \in F(\alpha, \beta)$, and $\beta \in F(\alpha, \beta)$. Hence

$$F(\alpha) \subseteq F(\alpha, \beta) \text{ and } \beta \in F(\alpha, \beta)$$

By definition,

$$(F(\alpha))(\beta) \subseteq F(\alpha, \beta).$$

Similarly,

$$F(\alpha) \subseteq (F(\alpha))(\beta) \text{ and } \beta \in (F(\alpha))(\beta).$$

Futhermore,

$$F \subseteq (F(\alpha))(\beta), \alpha \in (F(\alpha))(\beta \text{ and } \beta \in (F(\alpha)))(\beta).$$

We can conclude

$$F(\alpha, \beta) \subseteq (F(\alpha))(\beta)$$

by definition. □

Example 4.3.7. $\left[\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right) : \mathbb{Q}\right] = \left[\mathbb{Q}\left(\sqrt{2}, \sqrt{3}\right) : \mathbb{Q}(\sqrt{3})\right]\left[\mathbb{Q}\left(\sqrt{3}\right) : \mathbb{Q}\right]$

In general, we can extend this inductively. Let

$$K = F(\alpha_1, \alpha_2, ..., \alpha_k) = (F(\alpha_1, ..., \alpha_{k-1}))(\alpha_k) = F(\alpha_1, ..., \alpha_{k-2})(\alpha_{k-1})(\alpha_k)$$

which gives a sequence of fields:

$$F \subseteq F_0 \subseteq F_1 \subseteq ... \subseteq F_k = K$$

where $F_{i+1} = F_i(\alpha_{i+1})$ for $i = 0, ..., k-1$. If $\alpha_1, ..., \alpha_k$ are algebraic over F with degrees $n_1, ..., n_k$ respectively, then

$$[F_{i+1} : F_i] = [F_i(\alpha_{i+1}) : F_i] \leq [F_i(\alpha_{i+1}) : F] = n_{i+1}$$

with equality when $m_{\alpha_{i+1},F}(x)$ remains irreducible in $F_i[x]$. By the tower law,

$$[K : F] = [F_k : F_{k-1}]\,[F_{k-1} : F_{k-2}] \cdots [F_1 : F] \leq n_1 \cdot n_2 \cdots n_k.$$

We are able to use this to calculate basis for extensions $F(\alpha_1, ..., \alpha_k)$. We only need to examine this in the case $k = 2$:

$$F(\alpha, \beta) \text{ where } \alpha, \beta \text{ are algebraic.}$$

Consider $F(\alpha, \beta) = (F(\alpha))(\beta)$. We see the elements of $F(\alpha, \beta)$ can be written as

$$b_0 + b_1\beta + b_2\beta^2 + ... + \beta_{d-1}\beta^{d-1}$$

where $d = [F(\alpha, \beta) : F(\alpha)]$ and $b_j \in F(\alpha)$. Since $b_j \in F(\alpha)$, we have

$$b_j = a_{0j} + a_{1j}\alpha + a_{2j}\alpha^2 + ... + a_{(n-1)j}\alpha^{n-1}$$

where $a_{ij} \in F$ and $n = [F(\alpha) : F]$. Hence the elements of $F(\alpha, \beta)$ can be written in the form

$$\sum_{\substack{0 \leq i \leq n-1 \\ 0 \leq j \leq d-1}} a_{ij}\alpha^i\beta^j$$

Note . The elements $\alpha^i\beta^j$ are an F basis for $F(\alpha, \beta)$.

Example 4.3.8. $\mathbb{Q}(\sqrt{2},\sqrt{3})$.
From the tower law, we know

$$\Big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}\Big] = \Big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}(\sqrt{2})\Big]\Big[\mathbb{Q}(\sqrt{2}):\mathbb{Q}\Big] \leq 4.$$

This gives us two cases: $\big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}(\sqrt{2})\big] \in \{1,2\}$.
Consider $\big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}\sqrt{2}\big] = 1$. Then $\big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}\big] = 2 \implies \mathbb{Q}(\sqrt{2},\sqrt{3}) = \mathbb{Q}(\sqrt{2})$ so $\sqrt{3} \in \mathbb{Q}(\sqrt{2})$.
That is,

$$\sqrt{3} = a + b\sqrt{2} \text{ for some } a,b \in \mathbb{Q}$$

If $a = 0$ and $b \neq 0$, then

$$\begin{aligned} &\sqrt{3} = b\sqrt{2} \\ \implies &\sqrt{6} = 2b \\ \implies &\sqrt{6} \in \mathbb{Q}, \text{ a contradiction.} \end{aligned}$$

If $b = 0$ and $a \neq 0$, then $\sqrt{3} = 2a \implies \sqrt{3} \in \mathbb{Q}$, a contradiction.
Now, if $a, b \neq 0$ then

$$\begin{aligned} 3 = \left(a + b\sqrt{2}\right)^2 &= a^2 + 2b^2 + 2\sqrt{2}ab \\ \implies \frac{3 - a^2 - 2b^2}{2ab} &= \sqrt{2} \\ \implies &\sqrt{2} \in \mathbb{Q}, \text{ a contradiction.} \end{aligned}$$

Thus $\sqrt{3} \neq \mathbb{Q}(\sqrt{2}) \implies \big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}(\sqrt{2})\big] = 2$.
What's more: $x^2 - 3$ is irreducible over $\mathbb{Q}(\sqrt{2})$. Furthermore,

$$\Big[\mathbb{Q}(\sqrt{2},\sqrt{3}):\mathbb{Q}\Big] = 4$$

and the basis elements are $\sqrt{3}^i\sqrt{2}^j$ for $i,j = 0,1$. That is,

$$\mathbb{Q}(\sqrt{2},\sqrt{3}) = \left\{a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} : a,b,c,d \in \mathbb{Q}\right\}$$

Theorem 4.3.2. The extension K/F is finite if and only if K is generated by a finite number of algebraic elements over F. More precisely, a field generated by a finite number of algebraic elements of degree $n_1, n_2, ..., n_k$ is algebraic of degree at most $n_1 n_2 \cdots n_k$.

Proof. ($\Longleftarrow$) We already proved this direction.
($\Longrightarrow$) If K/F is finite of degree n, let $\alpha_1, ..., \alpha_n$ be a basis for K over F. We know $[F(\alpha_i) : F]$ divides $[KF]$ for $i = 1, 2, ..., n$. so each $[F(\alpha_i) : F]$ is finite and hence each α_i is algebraic over F. Since span $\{\alpha_1, \alpha_2, ..., \alpha_n\} = K$, K is generated by a finite number of algebraic elements over F. □

Corollary 4.3.3. Suppose α, β are algebraic over F. Then $\alpha \pm \beta, \alpha\beta, \alpha/\beta$ given $\beta \neq 0$ (in particular, β^{-1} for $\beta \neq 0$) are all algebraic.

Proof. Since $F(\alpha,\beta)$ is a field, $\alpha \pm \beta, \alpha\beta, \alpha/\beta \in F(\alpha)$. Since $F(\alpha,\beta)$ is generated by two algebraic numbers (a finite amount) over F, $F(\alpha,\beta)$ is finite and hence algebraic. So, $\alpha \pm \beta, \alpha\beta, \alpha/\beta$ are all algebraic. □

Corollary 4.3.4. Let L/F be an arbitrary extension. Then the collection of elements of L that are algebraic over F form a subfield of L.

Example 4.3.9. Consider $\mathbb{C}/\mathbb{Q}$ and let $\bar{\mathbb{Q}}$ denote the subfield of $\mathbb{C}$ containing all elements that are algebraic over $\mathbb{Q}$. We will show that $\bar{\mathbb{Q}}$ is an infinite algebraic extension of $\mathbb{Q}$.
Notice that $\sqrt[n]{2} \in \bar{\mathbb{Q}}$, $n > 1$ with any polynomial $x^n - 2$. By the tower law, we have

$$[\bar{\mathbb{Q}} : \mathbb{Q}] = [\bar{\mathbb{Q}} : \mathbb{Q}(\sqrt[n]{2})][\mathbb{Q}(\sqrt[n]{2} : \mathbb{Q})] \geq n \quad \forall n > 1$$

Hence $\bar{\mathbb{Q}}$ is an infinite extension of $\mathbb{Q}$. We call $\bar{\mathbb{Q}}$ the field of algebraic numbers.

4.4 Splitting Fields and Algebraic Closure

Alebraic extensions allow us to enlarge a field by adding roots of a polynomial equation, but sometimes one root is not enough. A polynomial may have a root in the extension, but its remaining roots may lie outside of the extension. For example, x^3-2 has a root, $\sqrt[3]{2}$, which appears in the extension $\mathbb{Q}(\sqrt[3]{2})$. However, the polynomial has two additional roots, $\sqrt[3]{2}\omega$ and $\sqrt[3]{2}\omega^2$ where ω is a primitive cube root of unity, which do not appear in $\mathbb{Q}(\sqrt[3]{2})$. Because of this, x^3-2 factors into $(x-\sqrt[3]{2})(x^2+\sqrt[3]{2}x+sqrt[3]4)$. Notice that the second factor is an irreducible quadratic over $\mathbb{Q}(\sqrt[3]{2})$. By enlarging the field to contain the remaining roots, we would be able to factor x^3-2 into linear factors, giving us a complete understanding of the polynomial. This motivates the following definition.

Definition 4.4.1. (Splitting Field)
the extension field K/F si called the splitting field for the polynomial $f(x)\in F[x]$ if $f(x)$ factors completely into linear factors in $K[x]$ and does not factor completely into linear factors over any proper subfield of K containing F.

Example 4.4.1. What is the splittinf field of $x^2-5\in\mathbb{Q}[x]$?

$$\mathbb{Q}(\sqrt{5}]$$

Note the two zeros of x^2-5, $\pm\sqrt{5}$, are in $\mathbb{Q}(\sqrt{5})$. This is the smallest field containing $\mathbb{Q}$ and the roots of x^2-5.

Remark 4.4.1. If $f(x)\in F[x]$ is of degree n, then $f(x)$ has at most n roots and has n roots, counting multiplicity, in F if and only if $f(x)$ splits completely over F.

Theorem 4.4.1. For any field F, if $f(x)\in F[x]$ then there exists an extension K/F which is a splitting field for $f(x)$.

Proof. Induction on the degree n of $f(x)$.

Base: $n=1$
$f(x)$ is linear, so we take $K=F$. Thus there exists a field containing in which $f(x)$ factors completely. In this case, $K=E=F$ since there is no proper subfield of F containing F.

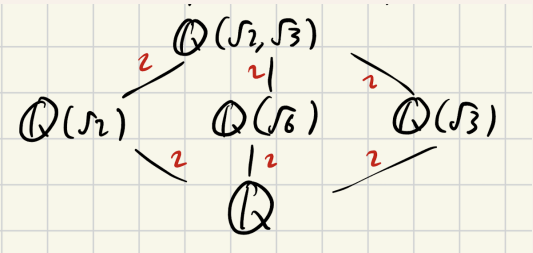
Inductive Step: $n>1$
Suppose now that $\deg(f(x))>1$. If the irreducible factors of $f(x)$ over F are all linear, then F is the splitting field of $f(x)$ so again we take $E=F$. Otherwise, at least one of the irreducible factors of $f(x)$ is of degree greater than 1 and at most n. There exists an extension E_1 of F where this irreducible factor, $p(x)$, has a zero. Letting this root be α, we then have

$$f(x)=(x-\alpha)g(x)$$

where $g(x)\in E_1[x]$ since $\deg(f(x))=n$. We know $\deg(g(x))=n-1$, so by induction there exists an extension E of E_1 which contains all the roots of $g(x)$. Since $\alpha\in E_1\subseteq E$, E contains all the roots of $f(x)$. To get the splitting field for $f(x)$, we let K be the interseciton of all subfields of E containing F which also contain all the roots of $f(x)$. Then K is the splitting field of $f(x)$. □

Definition 4.4.2. (Normal Extension)
If K is an algebraic extension of F which is the splitting field over F for a collection over F, then K is called a normal extension.

Example 4.4.2. Find the splitting field of $(x^2-2)(x^2-3)$. We feel strongly that $\mathbb{Q}(\sqrt{2},\sqrt{3})$ si the splitting field of this polynomial since $\pm\sqrt{2},\pm\sqrt{3}\in\mathbb{Q}(\sqrt{2},\sqrt{3})$. $\mathbb{Q}(\sqrt{2},\sqrt{3})$ contains $\mathbb{Q}$ and all the roots of $(x^2-2)(x^2-3)$, hence the splitting field for this polynomial over $\mathbb{Q}$, call it K, is contianed in $\mathbb{Q}(\sqrt{2},\sqrt{3})$. Furthermore, since $\mathbb{Q}\subseteq K$ and $\sqrt{2},\sqrt{3}\in K$, we have $\mathbb{Q}(\sqrt{2},\sqrt{3})\subseteq K$. This shows $K=\mathbb{Q}(\sqrt{2},\sqrt{3})$.
Consider the diagram of known subfields of $\mathbb{Q}(\sqrt{2},\sqrt{3})$. Recall the form of $\mathbb{Q}(\sqrt{2},\sqrt{3})=\{a+b\sqrt{2},c\sqrt{3}+d\sqrt{6}:a,b,c,d\in\mathbb{Q}\}$.



Example 4.4.3. What is the splitting field of x^3-2? It's not $\mathbb{Q}(\sqrt[3]{2})$. The roots of x^3-2 are $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, $\sqrt[3]{2}\omega^2$ for $\omega = \frac{-1}{2} + \frac{\sqrt{3}}{2}i$. So the splitting field for x^3-2 is $\mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2) = K$. Notice that since $\sqrt[3]{2}, \sqrt[3]{2}\omega \in K$, then $\frac{\sqrt[3]{2}\omega}{\sqrt[3]{2}} = \omega \in K$. Further, since $2, 1 \in K$ then $2\omega + 1 = \sqrt{-3} \in K$. Then $\mathbb{Q}(\sqrt{-3}, \sqrt[3]{2}) \subseteq K$. Furthermore, any field which contains $\sqrt{-3}$ and ω in turn contain all the roots of $x^3 - 2$, therefore

$$K \subseteq \mathbb{Q}(\sqrt{-3}, \sqrt[3]{2})$$

Let's use this to find the degree of K. We know

$$[\mathbb{Q}(\sqrt{-3}, \sqrt[3]{2}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{-3}, \sqrt[3]{2}) : \mathbb{Q}(\sqrt{-3}][\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] \leq 6$$

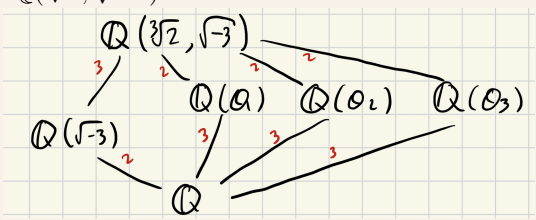
since $[\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] = 2$ and $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. We also know

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] \mid [\mathbb{Q}(\sqrt{-3}, \sqrt[3]{2}) : \mathbb{Q}]$$

and

$$[\mathbb{Q}(\sqrt{-3}) : \mathbb{Q}] \mid [\mathbb{Q}(\sqrt{-3}, \sqrt[3]{2}) : \mathbb{Q}]$$

Therefore, the degree of $\mathbb{Q}(\sqrt{-3}, \sqrt[3]{2})$ over $\mathbb{Q}$ is 6 (we have also shown that $x^3 - 2$ is irreducible over $\mathbb{Q}(\sqrt{-3})$). Known subfields of $\mathbb{Q}(\sqrt[3]{2}, \sqrt{-3})$:



where $\theta_1 = \sqrt[3]{2}, \theta_2 = \sqrt[3]{2}\omega, \theta_3 = \sqrt[3]{2}\omega^2$.

Proposition 4.4.1. A splitting field of a polynomial of degree n over F is of degree at most $n!$ over F.

Proof. If $f(x)$ is a polynomial of degree n over F, we can adjoin a root α to F, then the degree of $F(\alpha)$ over F is at most n. We can then factor $f(x)$ as

$$f(x) = (x - \alpha)g(x)$$

where $\deg(g(x)) = n - 1$. Hence, adjoining a root of $g(x)$ would have degree of at most $n - 1$ over $F(\alpha)$ and so on. □

4.5 Splitting Fields of Cyclotomic Fields

Consider the polynomial $x^n - 1 \in \mathbb{Q}[x]$. Consider the nth-roots of unity

$$U_n = \{\alpha \in \mathbb{C} : \alpha \text{ is a root of } x^n - 1\}$$

A nonzero complex number $a + bi \in \mathbb{C}$ can be written uniquely in the form

$$r(\cos\theta + i\sin\theta) = re^{i\theta}, \quad r > 0, \quad 0 \leq \theta < 2\pi$$

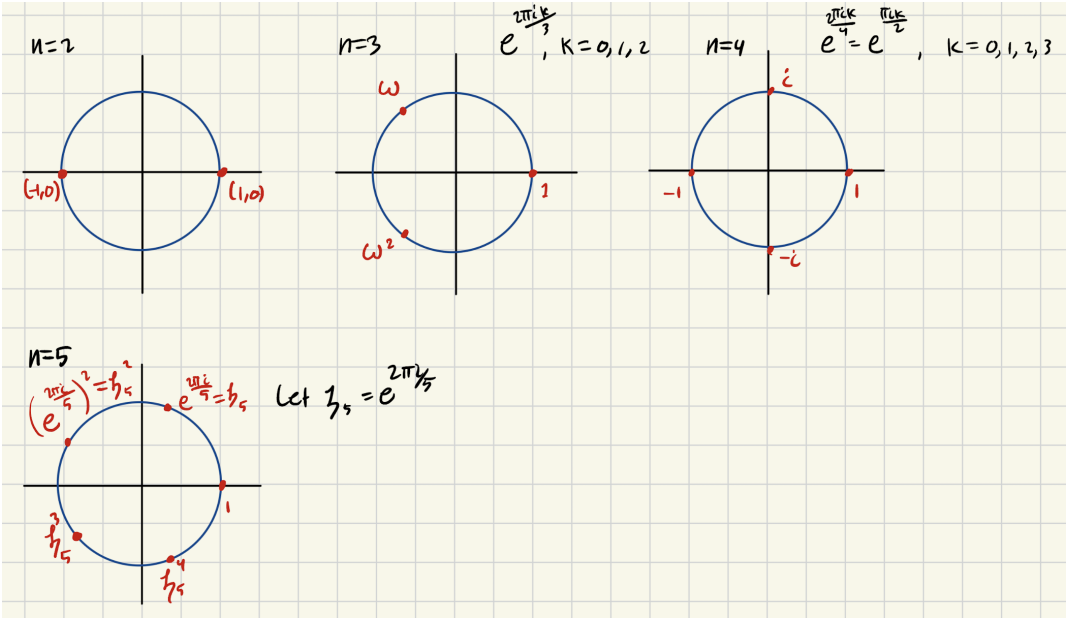
So, the roots of $x^n - 1$ are of the form

$$e^{\frac{2\pi ik}{n}} = \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right), \quad k = 0, 1, 2, ..., n-1$$

Notice that

$$\left(e^{\frac{2\pi ik}{n}}\right)^n = e^{2\pi ik} = \cos(2\pi k) + i\sin(2\pi k) = 1$$

So we have that $e^{\frac{2\pi ik}{n}}$ is a root of $x^n - 1$. Geometrically, $e^{\frac{2\pi ik}{n}}$ are equally spaced points on the unit circle in the complex plane starting at $(1, 0)$.



Since $x^n - 1$ has at most n roots in $\mathbb{C}$, it follows that all of the roots of $x^n - 1$ are of the form $e^{\frac{2\pi ik}{n}}$ for $k = 0, 1, ..., n-1$. It's pretty easy to show U_n is a group with respect to multiplication by constructing the following isomorphism:

$$\begin{aligned} \mathbb{Z}/n\mathbb{Z} &\to U_n \\ k &\mapsto \left(e^{\frac{2\pi i}{n}}\right)^k \end{aligned}$$

which would give us that $e^{\frac{2\pi i}{n}}$ is cyclic.

Definition 4.5.1. (Primitive Root of Unity)
A generator of U_n is called a primitive root of unity.

Let ζ be a primitive n^{th}-root of unity. Then

$$|\zeta|^k = \frac{n}{\gcd(n, k)}$$

Since $k = 0, 1, ..., n-1$, there are exactly $\phi(n)$ primitive n^{th}-roots of unity. Let $\zeta_n = e^{\frac{2\pi i}{n}}$. Some useful roots to

know:

$$\begin{aligned}
\zeta_1 &= 1 \\
\zeta_2 &= -1 \\
\zeta_3 &= -\frac{1}{2} + \frac{\sqrt{3}}{2}i \\
\zeta_4 &= i \\
\zeta_5 &= \frac{\sqrt{5}-1}{4} + \frac{\sqrt{10+2\sqrt{5}}}{4}i \\
\zeta_6 &= \frac{1}{2} + \frac{\sqrt{3}}{2}i \\
\zeta_7 &= e^{\frac{2\pi i}{7}} \\
\zeta_8 &= \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i
\end{aligned}$$

Definition 4.5.2. (Cyclotomic Field)
The field $\mathbb{Q}(\zeta_n)$ is called the cyclotomic field of nth-roots of unity.

Let p be a prime.

$$\begin{aligned}
[\mathbb{Q}(\zeta_p) : \mathbb{Q}] &= p-1 \\
x^p - 1 &= (x-1)(x^{p-1} + x^{p-2} + ... + x + 1) \\
\implies \frac{x^p-1}{x-1} &= x^{p-1} + x^{p-2} + ... + x + 1
\end{aligned}$$

We proved that $p(x) = \frac{x^p-1}{x-1}$ is irreducible ($p(x+1)$ with Eisenstein's Criterion). Since $\zeta_n \neq 1$, $n > 1$ where ζ_n is a root of $p(x)$, so $m_{\zeta_n,\mathbb{Q}}(x) = p(x)$. In general,

$$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n)$$

Example 4.5.1. Find splitting field of $x^p - 2 = f(x) \in \mathbb{Q}[x]$, where p is a prime. Note that the solutions to $f(x)$ are

$$\sqrt[p]{2}\zeta_p^k \text{ for } k = 0, 1, ..., p-1$$

Let K be the splitting field of $f(x)$ over $\mathbb{Q}$.

$$K = \mathbb{Q}\left(\sqrt[p]{2}, \sqrt[p]{2}\zeta_p, \sqrt[p]{2}\zeta_p^2, ..., \sqrt[p]{2}\zeta_p^{p-1}\right)$$

We know that $\sqrt[p]{2}, \sqrt[p]{2}\zeta_p \in K$. This implies

$$\frac{\sqrt[p]{2}\zeta_p}{\sqrt[p]{2}} = \zeta_p \in K$$

So, K contains $\mathbb{Q}, \sqrt[p]{2}, \zeta_p$, so it contains $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$. Note that since all the roots of $f(x)$ are of the form $\zeta_p^k \sqrt[p]{2}$, we have that all of $f(x)$'s roots are in $\mathbb{Q}(\sqrt[p]{2}, \zeta_p)$. Hence

$$K \subseteq \mathbb{Q}(\sqrt[p]{2}, \zeta_p)$$

So $K = \mathbb{Q}(\sqrt[p]{2}, \zeta_p)$ and

$$[K : \mathbb{Q}] = [K : \mathbb{Q}(\sqrt[p]{2})][\mathbb{Q}(\sqrt[p]{2}) : \mathbb{Q}] \leq (p-1)p$$

and $p \mid [K : \mathbb{Q}]$. Similarly, $p-1 \mid [K : \mathbb{Q}]$. Since $\gcd(p-1, p) = 1$, we have that $p(p-1) \mid [K : \mathbb{Q}]$. Hence, $[K : \mathbb{Q}] = p(p-1)$.

4.6 Derivative of a Polynomial

With splitting fields, we are able to factor polynomials into linear factors. Sometimes, these factors may repeat. For example, the polynomial $f(x) = (x-1)^2(x-2)$ has a repeated factor of $(x-1)$.

More generally, let F be a field and $f(x) \in F[x]$. For $f(x)$, we have

$$f(x) = (x-\alpha_1)^{n_1}(x-\alpha_2)^{n_2}\cdots(x-\alpha_k)^{n_k}$$

where each α_i are distinct elements of K, a splitting field for $f(x)$ and $n_i \geq 1 \forall i$. α_i is called a repeated root if $n_i > 1$ where n_i is the multiplcity of α_i.

In calculus, we know that if $f(x)$ has a repeated root at α, then $f'(\alpha) = 0$. We will make use of this idea to detect repeated roots of polynomials over fields.

Definition 4.6.1. (Derivative of a Polynomial)
Let $f(x) = \sum_{i=0}^{n} a_i x^i$ be a nonzero polynomial in $F[x]$. We define

$$D_x f(x) = \sum_{i=1}^{n} a_1 \cdot i x^{i-1} = a_1 + 2a_2 x + \cdots + n a_n x^{n-1}$$

to be the formal derivative of $f(x)$.

Definition 4.6.2. (Separable Polynomial)
A polynomial is separable if it has no multiple roots. A polynomial which is not separable is called inseparable.

Proposition 4.6.1. A polynomial $f(x)$ has a multiple root α if and only if α is also a root of $D_x f(x)$. I.e., $f(x)$ and $D_x f(x)$ are both divisible by the minimal polynomial of α over F. In particular, $f(x)$ is separable if and only if it is relatively prime to its derivative.

Example 4.6.1. The product rule, power rule and chain rule hold with formal derivatives. That is,

$$\begin{aligned} D_x\left(f(x)g(x)\right) &= \left(D_x f(x)\right)g(x) + f(x)\left(D_x g(x)\right) \\ D_x\left(\left(f(x)\right)^n\right) &= n\left(f(x)\right)^{n-1} D_x f(x) \\ D_x\left((x-\alpha)^n\right) &= n(x-\alpha)^{n-1} \end{aligned}$$

Example 4.6.2. The polynomial $x^{p^n} - x$ over $\mathbb{F}_p$ has derivative

$$p^n x^{p^n-1} - 1 = 0 - 1 = -1 \text{ (using characteristic } p)$$

so we can immediately see that $x^{p^n} - x$ is relatively prime to $D_x\left(x^{p^n-1} - x\right)$ and so $x^{p^n-1} - x$ is separable (it has no multiple roots).

Corollary 4.6.1. Every irreducible polynomial over a field of characteristic 0 is separable. A polynomial is separable if and only if it is the product of distinct irreducible polynomials.

Proof. Suppose $\text{char}(F) = 0$ and $p(x) \in F[x]$ is irreducible of degree n. Then

$$\deg\left(D_x\left(p(x)\right)\right) = n - 1$$

The only factors of the irreducible polynomial $p(x)$ are associates of $p(x)$ and associates of 1 so $D_x(p(x))$ must be relatively prime to $p(x)$. Hence, $p(x)$ is separable. □

Proposition 4.6.2. Let F be a field with $\text{char}(F) = p$ where p is prime. Then for any $a, b \in F$, $(a+b)^p = a^p + b^p$ and $(ab)^p = a^p b^p$. That is, the mapping defined by $\varphi(a) = a^p$ is an injective field homomorphism from F to F called the Frobenius automorphism.

Example 4.6.3. Here is an important example. From homework, we proved that if $K/\mathbb{F}_p$ is finite then $|K| = p^n$ for some n. We will fix some $n \in \mathbb{N}$ and show there exists an extension $F/\mathbb{F}_p$ such that $[F : \mathbb{F}_p] = n$.
Let $n > 0$ and consider $f(x) = x^{p^n} - x \in \mathbb{F}_p[x]$. Let K be the splitting field of $f(x)$ over $\mathbb{F}_p$. We know

$f(x)$ is separable, thus it has exactly p^n distinct roots. Let α, β be roots of $f(x)$. Then

$$\left.\begin{aligned}\alpha^{p^n} - \alpha = 0\\ \beta^{p^n} - \beta = 0\end{aligned}\right\} \implies \left.\begin{aligned}\alpha^{p^n} = \alpha\\ \beta^{p^n} = \beta\end{aligned}\right\}$$

Then

$$\begin{aligned}(\alpha\beta)^{p^n} &= \alpha^{p^n}\beta^{p^n} = \alpha\beta\\ (\alpha^{-1})^{p^n} &= (\alpha^{p^n})^{-1} = \alpha^{-1}\\ (\alpha+\beta)^{p^n} &= \alpha^{p^n} + \beta^{p^n} = \alpha + \beta\end{aligned}$$

Hence, the set of all p^n roots of $f(x)$, F, is closed under addition, multiplication, and inverse in K. That is, F is a subfield of K. Noting that $1 \in F$, we see $\mathbb{F}_p \subseteq K$. Hence $K \subseteq F$, so $K = F$. Since F is a vector space over $\mathbb{F}_p$ and $|F| = p^n$, it follows that $[F : \mathbb{F}_p] = n$. Hence, we have shown the existence of finite fields of order p^n.

Chapter 5

Galois Theory

5.1 Field Automorphisms and Galois Groups

After studying splitting fields, we are now in a position to study the symmetries of these fields. In particular, we will be interested in the field automorphisms of splitting fields. As we will see, automorphisms for splitting fields are closely tied to the roots of the polynomial that we are splitting. The collection of such automorphisms will form a group under composition, which will be known as the Galois group of the splitting field.

Definition 5.1.1. (Field Automorphism)
Let L be a field.

(i) A field isomorphism $\sigma : K \to K$ is called an automorphism of K. The set of all automorphisms of K is denoted by $\mathrm{Aut}(K)$.

(ii) We say $\sigma \in \mathrm{Aut}(K)$ fixes an element $\alpha \in K$ if $\sigma(\alpha) = \alpha$. If $F \subseteq K$, then σ is said to fix F if $\sigma(f) = f$ for all $f \in F$.

Example 5.1.1. $\sigma(a) = a \quad \forall a \in K$ is called the trivial automorphism (or the identity automorphism), denoted σ_1 or 1.

Note . The prime subfield of K, call it F (for now) is generated by 1 and since any automorphism τ, $\tau(1) = 1$, it follows that $\tau(a) = a$ for all $a \in F$. Hence any automorphisms of a field K fixes its prime subfield. So, all $\tau \in \mathbb{Q}(\sqrt{2})$ fix $\mathbb{Q}$.

Example 5.1.2. $\mathrm{Aut}(\mathbb{Q}) = \{\sigma_1\}$ since $\mathbb{Q}$ is its own prime subfield.

Definition 5.1.2. ($\mathrm{Aut}(K/F)$)
Let K/F be an extension of fields. Let $\mathrm{Aut}(K/F)$ be the collection of automorphisms of K which fix F.

Proposition 5.1.1. $\mathrm{Aut}(K)$ form a group with respect to composition and $\mathrm{Aut}(K/F)$ form a subgroup of $\mathrm{Aut}(K)$.

Proof. Composition is an associative binary operation on bijective morphisms. The identity of $\mathrm{Aut}(K)$ is σ_1. Isomorphisms have functional inverses, which are isomorphisms. So, $\mathrm{Aut}(K)$ forms a group. Now, let $\sigma, \tau \in \mathrm{Aut}(K/F)$. Then let $a \in F$. We have

$$\begin{aligned} \sigma\tau^{-1}(a) &= \sigma\tau^{-1}(\tau(a)) \\ &= \sigma(a) \\ &= a \end{aligned}$$

Hence, $\mathrm{Aut}(K/F) \leq \mathrm{Aut}(K)$. □

Proposition 5.1.2. Let K/F be a field extension and let $\alpha \in K$ be algebraic over F. Then for any $\sigma \in \mathrm{Aut}(K/F)$, $\sigma(\alpha)$ is a root of the minimal polynomial for α over F. I.e., $\mathrm{Aut}(K/F)$ permutes the roots of irreducible polynomials. Equivalently, any polynomial in $F[x]$ which has α as a root will also have $\sigma(\alpha)$ as a root.

Proof. If α satisfies $\sum_{i=0}^{\infty} a_i \alpha^i = 0$ where $a_i \in F$, then

$$\begin{aligned} \sigma\left(\sum_{i=0}^{\infty} a_i \alpha^i\right) &= \sigma(0) \\ \implies \sum_{i=0}^{\infty} \sigma(a_i \alpha^i) &= 0 \\ \implies \sum_{i=0}^{\infty} \sigma(a_i)\sigma(\alpha^i) &= 0 \\ \implies \sum_{i=0}^{\infty} a_i \sigma(\alpha^i) &= 0 \end{aligned}$$

□

Example 5.1.3. Consider $K = \mathbb{Q}(\sqrt{2})$. If $\tau \in \mathrm{Aut}(K) = \mathrm{Aut}(K/Q)$, then $\tau(\sqrt{2}) = \pm\sqrt{2}$ since $\pm\sqrt{2}$ are the two roots of $m_{\sqrt{2},\mathbb{Q}}(x)$. We have two possible automorphisms:

$$\begin{aligned} 1 &: \sqrt{2} \mapsto \sqrt{2} \\ \sigma &: \sqrt{2} \mapsto -\sqrt{2} \end{aligned}$$

Note σ^2:

$$\begin{aligned} \sigma^2(\sqrt{2}) &= \sigma(-\sqrt{2}) \\ &= -\sigma(\sqrt{2}) \\ &= -(-\sqrt{2}) \\ &= \sqrt{2} \\ \implies \sigma^2 &= 1 \end{aligned}$$

Then $\mathrm{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \langle\sigma\rangle \cong \mathbb{Z}/2\mathbb{Z}$.

Example 5.1.4. Consider $\mathbb{Q}(\sqrt[3]{2})$. If $\tau \in \mathrm{Aut}(K)$, we have the following possibilities: $\tau \in (\sqrt[3]{2}) \in \left\{\sqrt[3]{2}, \sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2\right\}$ since $m_{\sqrt[3]{2},\mathbb{Q}}(x) = x^3 - 2$. Since $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, τ cannot map to $\sqrt[3]{2}\omega, \sqrt[3]{2}\omega^2 \in \mathbb{C}$. Hence $\tau(\sqrt[3]{2}) = \sqrt[3]{2}$ ($\tau = 1$). Thus there is only one automorphism of $\mathbb{Q}(\sqrt[3]{2})$, namely 1. Hence, $\mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}) = 1$.

Comparing $\mathrm{Aut}(\mathbb{Q}(\sqrt{2}))$ and $\mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2}))$, we see that $\mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2}))$ is missing clear automorphisms. In general, we see that if K/F is generated by some collection of elements, then $\sigma \in \mathrm{Aut}(K/F)$ is completely determined by where σ maps the generators.

Proposition 5.1.3. Let $H \leq \mathrm{Aut}(K)$. Then the elements of K fixed by all automorphisms in H is a subfield of K.

Proof. Let $h \in H$, $a, b \in F$. Then, by definition, $h(a) = a$ and $h(b) = b$ so that

$$h(a \pm b) = h(a) \pm h(b).$$

Further,

$$h(ab) = h(a)h(b)$$

and

$$h(a^{-1}) = h(a)^{-1} = a^{-1}$$

Thus, F is a field. □

Note . H does not have to be a subgroup for the proposition to hold.

We the above proposition, we take interest in the elements in K that are fixed by the automorphisms of K.

Definition 5.1.3. (Fixed Field)
If $H \leq \text{Aut}(K)$, the subfield fixed by H is called the fixed field of H.

Proposition 5.1.4. The association of groups of $\text{Aut}(K)$ to fixed fields and fixed fields to subgroups is inclusion reversing. Namely,

(i) If $F_1 \subseteq F_2 \subseteq K$ are subfields of K, then $\text{Aut}(K/F_2) \subseteq \text{Aut}(K/F_1)$.

(ii) If $H_1 \leq H_2 \leq \text{Aut}(K)$ where F_1 and F_2 are the associated fixed fields of H_1 and H_2, respectively, then $F_1 \subseteq F_2$.

Proof. (i) Let $\sigma \in \text{Aut}(K/F_2)$. Let $a \in F_1$. Then $a \in F_2$ so

$$\sigma(a) = a$$

Hence, σ fixes F_1 and

$$\text{Aut}(K/F_2) \subseteq \text{Aut}(K/F_1)$$

(ii) Let $\sigma \in H_1$. Let $a \in F_1$. Since $H_1 \subseteq H_2$, $\sigma \in H_2$ so

$$a = \sigma(a) \in F_2$$

Example 5.1.5. Let $K = \mathbb{Q}(\sqrt{2})$. Then the fixed field of

$$\text{Aut}(\mathbb{Q}(\sqrt{2})) = \text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{1, \sigma\}$$

is the set of all $a + b\sqrt{2} \in \mathbb{Q}(\sqrt{2})$ such that

$$\begin{aligned} 1(a + b\sqrt{2}) &= a + b\sqrt{2} \\ \sigma(a + b\sqrt{2}) &= a + b\sqrt{2} \end{aligned}$$

Thus, we must have that

$$\begin{aligned} a - b\sqrt{2} &= a + b\sqrt{2} \\ \implies -b\sqrt{2} &= b\sqrt{2} \\ \implies -b &= b \\ \implies b &= 0 \end{aligned}$$

So, the fixed field of $\text{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$ is just $\mathbb{Q}$.

Example 5.1.6. Let $K = \mathbb{Q}(\sqrt{2})$. The fixed field of $\text{Aut}(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}) = 1$ is $Q(\sqrt[3]{2})$. Notice that for $\mathbb{Q}(\sqrt{2})$ we have

Subgroups of Aut$(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$:		**Fixed fields:**
$\{1\}$	$\longleftrightarrow$	$\mathbb{Q}(\sqrt{2})$
$\{1, \sigma\}$	$\longleftrightarrow$	$\mathbb{Q}$

With $\mathbb{Q}(\sqrt[3]{2})$, we have

Subgroups of Aut$(\mathbb{Q}(\sqrt{2})/\mathbb{Q})$:		**Fixed fields:**
$\{1\}$	$\longleftrightarrow$	$\mathbb{Q}(\sqrt[3]{2})$

but we have no set of automorphisms which fix $\mathbb{Q}$.

Proposition 5.1.5. Let E be the splitting field over F of some polynomial $f(x) \in F[x]$. Then

$$|\text{Aut}(E/F)| \leq [E : F]$$

with equality when $f(x)$ is separable over F.

Definition 5.1.4. (Galois Extensions and Groups)
Let K/F be a finite extension. Then K is said to be Galois over F (K/F is a Galois Extension) if $|\mathrm{Aut}(K/F)| = [K : F]$. If K/F is Galois, the group of automorphisms $\mathrm{Aut}(K/F)$ is called the Galois group of K/F, denoted $\mathrm{Gal}\,(K/F)$.

Corollary 5.1.1. If K is the splitting field over F of a separable polynomial $f(x)$, then K/F is Galois.

Definition 5.1.5. (Galois Group of a Polynomial)
If $f(x)$ is a separable polynomial over F, then the Galois group of $f(x)$ over F is $\mathrm{Gal}\,(K/F)$ where K is the splitting field of $f(x)$ over F.

Example 5.1.7. Show $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is Galois.
We know that $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$ and that $\mathrm{Aut}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{1, \sigma\}$. Hence, $\mathbb{Q}(\sqrt{2})/\mathbb{Q}$ is a Galois extension. Alternatively, we know that $\mathbb{Q}(\sqrt{2})$ is the splitting field of $x^2 - 2 \in \mathbb{Q}[x]$. Hence, $\mathbb{Q}(\sqrt{2})$ is Galois over $\mathbb{Q}$.

Example 5.1.8. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ is not Galois since

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3 \text{ but } \mathrm{Aut}(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}) = 1$$

Example 5.1.9. The extension $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ is Galois since it is the splitting field of the separable polynomial $(x^2 - 2)(x^2 - 3) \in \mathbb{Q}[x]$. Calculating the Galois group:
Any $\sigma \in \mathrm{Gal}\left(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}\right)$ can be determined by its action on $\sqrt{2}$ and $\sqrt{3}$. We know that $\sqrt{2}$ and $\sqrt{3}$ must map to $\pm\sqrt{2}$ and $\pm\sqrt{3}$, respectively. Hence, the only possible automorphisms are

$$\begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases} \quad \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$$
$$\begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases} \quad \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$$

Let

$$\sigma = \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}, \quad \tau = \begin{cases} \sqrt{2} \mapsto \sqrt{2} \\ \sqrt{3} \mapsto \sqrt{3} \end{cases}$$

Notice

$$\sigma^2(\sqrt{2}) = \sqrt{2} \text{ and } \sigma^2(\sqrt{3}) = \sqrt{3} \implies \sigma^2 = 1$$

Similarly, $\tau^2 = 1$. Furthermore,

$$\sigma(\tau(\sqrt{2})) = -\sqrt{2}, \quad \sigma(\tau(\sqrt{3})) = -\sqrt{3}$$

so

$$\sigma\tau = \begin{cases} \sqrt{2} \mapsto -\sqrt{2} \\ \sqrt{3} \mapsto -\sqrt{3} \end{cases}$$

Lastly, $\sigma\tau = 1$ so

$$\mathrm{Gal}\left(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}\right) \cong \mathbb{Z}_2 \times \mathbb{Z}_2$$

Example 5.1.10. Let's look at the corresponding fixed fields of the previous example.

Subgroups of Gal $\left(\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}\right)$:		**Corresponding fixed fields:**
1	$\longleftrightarrow$	$\mathbb{Q}(\sqrt{2}, \sqrt{3})$
$\langle\sigma\rangle$	$\longleftrightarrow$	$\mathbb{Q}(\sqrt{3})$
$\langle\tau\rangle$	$\longleftrightarrow$	$\mathbb{Q}(\sqrt{2})$
$\langle\sigma\tau\rangle$	$\longleftrightarrow$	$\mathbb{Q}(\sqrt{6})$
$\{1, \sigma, \tau, \sigma\tau\}$	$\longleftrightarrow$	$\mathbb{Q}$

To find the fixed field for $\langle\sigma\rangle$, note that

$$\begin{aligned}\sigma(a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}) &= a+b\sigma(\sqrt{2})+c\sigma(\sqrt{3})+d\sigma(\sqrt{6})\\ &= a-b\sqrt{2}+c\sqrt{3}+d\sigma(\sqrt{2})\sigma(\sqrt{3})\\ &= a-b\sqrt{2}+c\sqrt{3}-d\sqrt{6}\end{aligned}$$

An element in $\mathbb{Q}(\sqrt{2},\sqrt{3})$ is fixed by σ if

$$\begin{aligned}a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6} &= a-b\sqrt{2}+c\sqrt{3}-d\sqrt{6}\\ \implies b &= -b \text{ and } d=-d\\ \implies b=d &= 0\end{aligned}$$

so $\sigma(a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}) = a+c\sqrt{3}\in\mathbb{Q}(\sqrt{3})$.
Similarly, for $\langle\tau\rangle$,

$$\begin{aligned}\tau(a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}) &= a+b\sqrt{2}-c\sqrt{3}-d\sqrt{6}\\ \implies c &= 0 = d\\ \implies \tau(a+b\sqrt{2}+c\sqrt{3}+d\sqrt{6}) &= a+b\sqrt{2}\in\mathbb{Q}(\sqrt{2})\end{aligned}$$

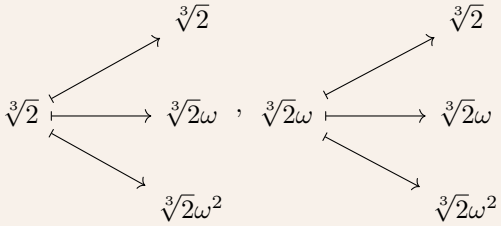
Example 5.1.11. Let $p(x)=x^3-2\in\mathbb{Q}[x]$. We have found the splitting field of $p(x)$ over $\mathbb{Q}$ to be $\mathbb{Q}(\sqrt[3]{2},\omega)$ where $\omega=\frac{-1+\sqrt{3}i}{2}$.
Notice that x^3-2 has three distinct roots, so $p(x)$ is separable and so its splitting field is Galois over $\mathbb{Q}$. We also know

$$[\mathbb{Q}(\sqrt[3]{2},\omega):\mathbb{Q}] = 6 = \left|\mathrm{Gal}\left(\mathbb{Q}(\sqrt[2]{3},\omega)/\mathbb{Q}\right)\right|.$$

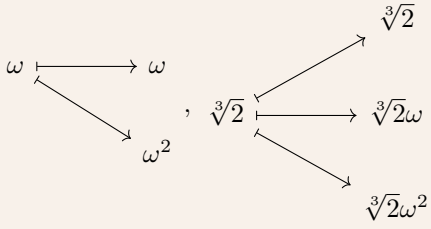
Suppose we write the splitting field as

$$\mathbb{Q}(\sqrt[3]{2},\sqrt[3]{2}\omega)$$

We know that any automorphism will send roots of minimal polynomials to other roots. Written this way, we may have



This gives $3\cdot 3=9$ possibilities for automorphisms, which is larger than the order of the Galois group. Using $\mathbb{Q}(\sqrt[3]{2},\omega)$, we have a more compatible number of potential automorphisms. Any automorphism can send



($m_{\omega,\mathbb{Q}}(x)=x^2+x+1$, $m_{\sqrt[3]{2},\mathbb{Q}}(x)=x^3-2$). This gives six automorphisms, which is exactly the size of the Galois group. Let

$$1:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\\ \omega\mapsto\omega\end{cases}$$

$$\sigma:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\omega\\ \omega\mapsto\omega\end{cases}\qquad\qquad \tau:\begin{cases}\sqrt[3]{2}\mapsto\sqrt[3]{2}\\ \omega\mapsto\omega^2=-1-\omega\end{cases}$$

$$\sigma^2 : \begin{cases} \sqrt[3]{2} \mapsto \sigma(\sqrt[3]{2}\omega) \mapsto \sqrt[3]{2}\omega^2 = \sqrt[3]{2}(-1-\omega) \\ \omega \mapsto \omega \end{cases} \qquad \tau^2 : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2} \\ \omega \mapsto \left(\omega^2\right)^2 = \omega^4 = \omega \end{cases}$$

$$\sigma^3 : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\omega(-1-\omega) = \sqrt[3]{2}(-\omega+1+\omega) = \sqrt[3]{2} \\ \omega \mapsto \omega \end{cases}$$

Notice that $\sigma^3 = 1 = \tau^2$, so $|\sigma| = |\langle\sigma\rangle| = 3$ and $|\tau| = |\langle\tau\rangle| = 2$.

$$\sigma\tau : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\omega \\ \omega \mapsto -1-\omega \end{cases} \qquad \tau\sigma : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}(-1-\omega) \\ \omega \mapsto (-1-\omega) \end{cases}$$

$\omega\tau \neq \tau\omega$, so the Galois group is not abelian.

$$\tau\sigma^2 : \begin{cases} \sqrt[3]{2} \mapsto \sqrt[3]{2}\omega \\ \omega \mapsto -1-\omega \end{cases}$$

$$\sigma\tau = \tau\sigma^2 \iff \sigma\tau = \tau\sigma^{-1}$$

Hence,

$$\mathrm{Gal}\left(\mathbb{Q}(\sqrt[3]{2},\omega)/\mathbb{Q}\right) \cong \left\langle \sigma,\tau : \sigma^3 = \tau^2 = 1, \sigma\tau = \tau\sigma^{-1} \right\rangle \cong D_6$$

Example 5.1.12. Let's compute the fixed field of $\langle\sigma\rangle$ from the previous example.
We need to see what is fixed by the generator σ. Lets look at the basis for $\mathbb{Q}(\sqrt[3]{2},\omega)$:

$$\left\{1, \sqrt[3]{2}, \left(\sqrt[3]{2}\right)^2, \omega, \sqrt[3]{2}\omega, \left(\sqrt[3]{2}\right)^2\omega\right\}.$$

Then,

$$\begin{aligned} \sigma(1) &= 1 \\ \sigma\left(\sqrt[3]{2}\right) &= \sqrt[3]{2}\omega \\ \sigma\left((\sqrt[3]{2})^2\right) &= \sigma\left(\sqrt[3]{2}\right)\sigma\left(\sqrt[3]{2}\right) = \left(\sqrt[3]{2}\omega\right)\left(\sqrt[3]{2}\omega\right) = \sqrt[3]{4}(-1-\omega) \\ \sigma(\omega) &= \omega \\ \sigma\left(\sqrt[3]{2}\omega\right) &= \sigma\left(\sqrt[3]{2}\right)\sigma(\omega) = \left(\sqrt[3]{2}\omega\right)(\omega) = \sqrt[3]{2}(-1-\omega) \\ \sigma\left((\sqrt[3]{2})^2\omega\right) &= \sigma\left((\sqrt[3]{2})^2\right)\sigma(\omega) = \left(\sqrt[3]{4}(-1-\omega)\right)(\omega) = \sqrt[3]{4}(-\omega+1+\omega) = \sqrt[3]{4} \end{aligned}$$

So,

$$\begin{aligned} \sigma\left(a + b\sqrt[3]{2} + c(\sqrt[3]{2})^2 + d\omega + e\sqrt[3]{2}\omega + f(\sqrt[3]{2})^2\omega\right) &= a + b\sqrt[3]{2}\omega + c\left((\sqrt[3]{2})^2\omega^2\right) + dw + e\sqrt[3]{2}\omega^2 + f(\sqrt[3]{2})^2 \\ &= a + b\sqrt[3]{2}\omega - c(\sqrt[3]{2})^2(1+\omega) + d\omega - e\sqrt[3]{2}(1+w) + f(\sqrt[3]{2})^2 \\ &= a - e\sqrt[3]{2} + (f-c)(\sqrt[3]{2})^2 + d\omega + (b-e)\sqrt[3]{2}\omega - c(\sqrt[3]{2})^2\omega \end{aligned}$$

For our element to be fixed, we need

$$\begin{aligned} a &= a & \qquad f &= -c \\ b &= -e & \qquad c &= f - c \\ d &= d & \qquad e &= b - e \end{aligned}$$

We have

$$e = b - e = -e - e \implies e = 0 \implies b = 0$$

Similarly,

$$c = 0 \implies f = 0$$

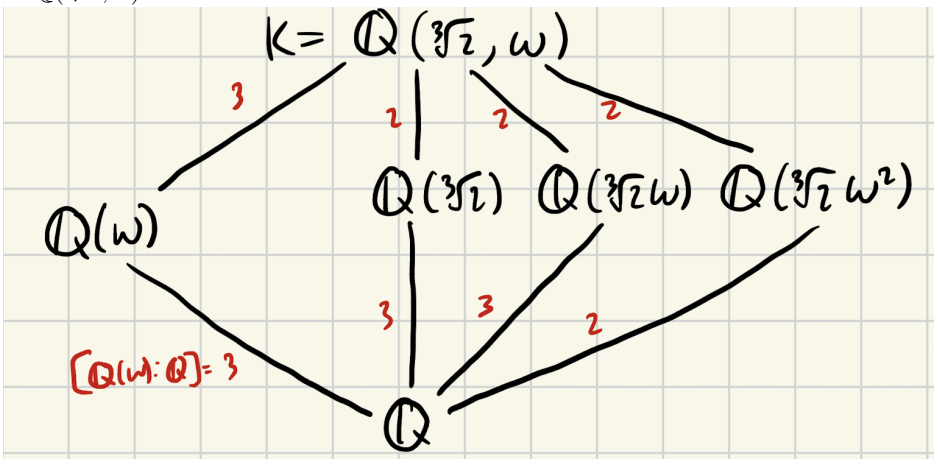
Hence, any elements fixed by σ are of the form $a + dw$. The fixed field of $\langle\sigma\rangle$ is $\mathbb{Q}(\omega)$.

Example 5.1.13. $\mathbb{Q}(\sqrt{2}, \sqrt[3]{3}, \sqrt[5]{5}, ...) = \bigcap_{\substack{F \supseteq \mathbb{Q} \\ \sqrt[p_i]{p_i} \in F}} F$ Note that

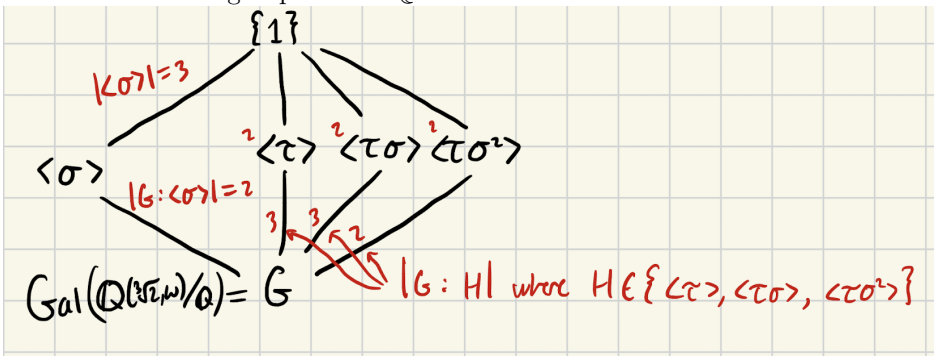
$$\bigcap_{\substack{F \supseteq \mathbb{Q} \\ \sqrt[p_i]{p_i} \in F}} F = \bigcup_{i=1}^{\infty} K_i \text{ where } K_i = \mathbb{Q}\left(\sqrt[2]{2}, \sqrt[3]{3}, \sqrt[5]{5}, ..., \sqrt[p_i]{p_i}\right)$$

5.2 The Fundamental Theorem of Galois Theory

Previously, we found the splitting field for $x^3 - 2$, $K = \mathbb{Q}(\sqrt[3]{2}, \omega)$, and we calculated $\mathrm{Gal}\left(\mathbb{Q}(\sqrt[3]{2}, \omega)/\mathbb{Q}\right)$. We also looked at subfields of $\mathbb{Q}(\sqrt[3]{2}, \omega)$:



The subgroup lattice of the Galois group K over $\mathbb{Q}$:



Theorem 5.2.1. (The Fundamental Theorem of Galois Theory)
Let K/F be a Galois extension and set $G = \mathrm{Gal}\,(K/F)$. Then there is a bijection between the set of subfields E of K containing F and the subgroups of G

$$\begin{array}{ccc} K & & 1 \\ | & & | \\ E & \longleftrightarrow & H \\ | & & | \\ F & & G \end{array}$$

given by the correspondence

$$E \to \{\text{the elements of } G \text{ fixing } E\}$$
$$\{\text{The fixed field of } H\} \leftarrow H$$

which are inverses to each other. Under this correspondence, we have the following results:

(i) (Inclusion Reversing) If E_1, E_2 correspond to H_1 and H_2, respectively, then

$$E_1 \subseteq E_2 \iff H_2 \subseteq H_1$$

(ii) $[K : E] = |H|$ and $[E : F] = |G : H|$

$$\begin{array}{c} K \\ \Big| \; \} |H| \\ E \\ \Big| \; \} |G:H| \\ F \end{array}$$

(iii) The extension K/E is always Galois with $\mathrm{Gal}\,(K/E) = H$.

$$\begin{array}{c} K \\ | \quad H \\ E \end{array}$$

(iv) E is Galois over F if and only if H is a normal subroup of G. If this is the case, then the Galois group of E over F is isomorphic to the quotient group G/H. That is,

$$\mathrm{Gal}\,(E/F) \cong G/H$$

(v) If E_1, E_2 correspond to H_1, H_2, respectively, then $E_1 \cap E_2$ corresponds to the group $\langle H_1, H_2 \rangle$ and E_1E_2 corresponds to $H_1 \cap H_2$.

Example 5.2.1. Let $p(x) = x^8 - 2$. Find the Galois group of the splitting field of $p(x)$ over $\mathbb{Q}$. The zeros of $p(x)$ are:

$$\theta, \theta\zeta, \theta\zeta^2, \theta\zeta^3, ..., \theta\zeta^7 \text{ where } \zeta = \zeta_8, \quad \theta = \sqrt[8]{2}$$

Hence, $\theta, \theta\zeta \in K$. Thus

$$\zeta = \frac{\theta\zeta}{\theta} \in K$$
$$\implies \mathbb{Q}(\theta, \zeta) \subseteq K$$

Further, each root of $p(x)$ can be constructed with θ and ζ. Hence, $\theta\zeta^n \in \mathbb{Q}(\theta, \zeta)$ for all $n = 0, 1, ..., 7$. This gives $K \subseteq \mathbb{Q}(\theta, \zeta)$. Hence, $K = \mathbb{Q}(\theta, \zeta)$. We know $\zeta = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$, thus

$$\mathbb{Q}(\zeta) = \mathbb{Q}(i, \sqrt{2})$$
$$\left(\zeta + \zeta^7 = \sqrt{2}, \quad \zeta^2 = i\right)$$

So, $K = \mathbb{Q}(\theta, i, \sqrt{2})$. But $\theta^4 = \sqrt{2}$, so $K = \mathbb{Q}(\theta, i)$ since $x^8 - 2$ is irreducible over $\mathbb{Q}$ by Eisenstein's. We have

$$[\mathbb{Q}(\theta) : \mathbb{Q}] = 8$$

and since $x^2 + 1$ is irreducible over $\mathbb{Q}$ with roots $\pm i$

$$[\mathbb{Q}(i) : \mathbb{Q}] = 2$$

so

$$[K : \mathbb{Q}] \leq 2 \cdot 8 = 16$$
$$[K : \mathbb{Q}] = [K : \mathbb{Q}(\theta)][\mathbb{Q}(\theta) : \mathbb{Q}]$$

However, $K \neq \mathbb{Q}(\theta)$ since $\mathbb{Q}(\theta) \subseteq \mathbb{R}$ but $i \notin \mathbb{R}$. Thus

$$[K : \mathbb{Q}(\theta)] = 2$$

It follows that $[K : \mathbb{Q}] = 16$. Thus $G = \mathrm{Gal}\,(K/\mathbb{Q})$ has order 16. The possible automorphisms are

$$\begin{cases} \theta \mapsto \theta\zeta^a \\ i \mapsto \pm i \end{cases} \text{ where } a = 0, 1, ..., 7$$

Define

$$\sigma : \begin{cases} \theta \mapsto \theta\zeta \\ i \mapsto i \end{cases} \qquad \tau : \begin{cases} \theta \mapsto \theta \\ i \mapsto -i \end{cases}$$
$$(\zeta \mapsto \zeta^5) \qquad (\zeta \mapsto \bar{\zeta} = \zeta^7) \qquad (*)$$
$$\sigma^2 : \begin{cases} \theta \mapsto \sigma(\theta\zeta) = \theta\zeta \cdot \theta^5 = \theta\zeta^6 \\ i \mapsto i \end{cases}$$
$$\vdots$$

Continuing, we get $|\sigma| = 8$ and $|\tau| = 2$. The group is

$$\left\{1, \sigma, \sigma^2, ..., \sigma^7, \tau, \tau\sigma, ..., \tau\sigma^7\right\}$$

with $\sigma\tau = \tau\sigma^3$. Thus

$$G = \left\langle \tau, \sigma : \tau^2 = \sigma^8 = 1, \sigma\tau = \tau\sigma^3 \right\rangle$$

This is a quasi-dihedral group.
$(*)$ Notice

$$\begin{aligned}
\sigma(\sqrt{2}) &= \sigma(\theta^4) \\
&= \sigma(\theta)^4 \\
&= (\theta\zeta)^4 \\
&= \theta^4\zeta^4 \\
&= \theta^4(-1) \\
&= -\sqrt{2}
\end{aligned}$$

and so

$$\zeta = \frac{1+i}{\sqrt{2}}, \quad \sigma(\zeta) = \frac{1+i}{-\sqrt{2}} = -\zeta = \zeta^5$$

5.3 Cyclotomic Extensions

Consider $\mathbb{Q}(\zeta_n)$, where $\zeta_n = e^{\frac{2\pi i}{n}}$. We know:

(i) $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n)$.

(ii) $\mathbb{Q}(\zeta_n)$ is the splitting field of $\Phi_n(x) = \prod_{1 \le a \le n} (x - \zeta_n^a)$ where $\gcd(a, n) = 1$.

By construction, Φ_n is separable of degree $\phi(n)$. Thus, $\mathbb{Q}(\zeta_n)$ is Galois over $\mathbb{Q}$. Any automorphism in $\mathrm{Gal}\,(\mathbb{Q}(\zeta_n)/\mathbb{Q}) = G$ is completely determined by where it maps ζ_n. Hence, if $\sigma \in G$, then $\sigma(\zeta_n) = \zeta_n^a$ where $1 \le a \le n$ with $\gcd(a, n) = 1$. This gives $\phi(n)$ possible mappings which is exactly $|G| = [\mathbb{Q}(\zeta_n) : \mathbb{Q}]$. For each $1 \le a \le n$ with $\gcd(a, n) = 1$, define $\sigma_{a \bmod n}$ to be the automorphism of $\mathbb{Q}(\zeta_n)$ such that

$$\zeta_n \to \zeta_n^a$$

Theorem 5.3.1. The Galois group of $\mathbb{Q}(\zeta_n)$ is isomorphic to $(\mathbb{Z}/nZ)^\times$, the group of units modulo n. The isomorphism is given by the map

$$\begin{aligned} (\mathbb{Z}/nZ)^\times &\to \mathbb{Q}(\zeta_n) \\ a(\bmod n) &\mapsto \sigma_{a(\bmod n)} \end{aligned}$$

where the σ_a is defined as above.

Proof. The discussion above shows that σ_a is an automorphism of $\mathbb{Q}(\zeta_n)$ for any a with $\gcd(a, n) = 1$. We will show this is a homomorphism:

$$\sigma_a\sigma_b(\zeta_n) = \sigma_a(\zeta_n^b) = (\zeta_n^b)^a = \zeta_n^{ab} = \sigma_{ab}(\zeta_n)$$

Thus, $\sigma_a\sigma_b = \sigma_{ab}$, so our mapping is a homomorphism. We know this is bijective since every automorphism of $\mathbb{Q}(\zeta_n)$ is of the form σ_a and each σ_a is mapped to and from a distinct a mod n. □

Example 5.3.1. The field $\mathbb{Q}(\zeta_5)$ is a Galois extension of $\mathbb{Q}$ with

$$G = \mathrm{Gal}\,(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong (\mathbb{Z}/5\mathbb{Z})^\times \cong \mathbb{Z}/4\mathbb{Z}$$

($\mathbb{Z}_n^\times$ is cyclic when $n = 2, 4, p^\alpha, 2p^\alpha$ where p is an odd prime). Hence, $G = \{\sigma_1 = 1, \sigma_2, \sigma_3, \sigma_4\}$. Let's find the order of σ_2:

$$\begin{aligned} &\sigma_2 : \zeta_5 \mapsto \zeta_5^2 \\ &\sigma_2^2 : \zeta_5 \mapsto \zeta_5^4 \implies \sigma_2^2 = \sigma_4 \\ &\sigma_2^3 : \zeta_5 \mapsto \zeta_5^8 \equiv \zeta_5^3 \implies \sigma_2^3 = \sigma_3 \\ &\sigma_2^4 : \zeta_5 \mapsto \zeta_5^6 \equiv \zeta_5 \implies \sigma_2^4 = \sigma_1 \end{aligned}$$

Thus $|\sigma_2| = 4$. What are the subfields of $\mathbb{Q}(\zeta_5)$ which contain $\mathbb{Q}$?

$$\begin{array}{ccc} \langle\sigma_1\rangle & & \mathbb{Q}(\zeta_5) \\ | & & | \\ \langle\sigma_4\rangle & & ? \\ | & & | \\ \mathrm{Gal}\,(\mathbb{Q}(\zeta_5)/\mathbb{Q}) \cong \mathbb{Z}/4\mathbb{Z} & & \mathbb{Q} \end{array}$$

Let $\alpha = \zeta_5 + \sigma_4(\zeta_5) = \zeta_5 + \zeta_5^4 = \zeta_5 + \zeta_5^{-1}$. Then

$$\begin{aligned} \sigma_4(\alpha) &= \sigma_4(\zeta_5 + \zeta_5^{-1}) \\ &= \sigma_4(\zeta_5) + \sigma_4(\zeta_5^{-1}) \\ &= \zeta_5^4 + \zeta_5^{-4} \\ &= \zeta_5^{-1} + \zeta_5. \end{aligned}$$

α is fixed by σ_4, so α is in the fixed field associated with $\langle\sigma_4\rangle$. We know $[\mathbb{Q}(\zeta_5) : \mathbb{Q}] \neq 1$ (basis for $\mathbb{Q}(\zeta_5)$ is $1, \zeta_5, \zeta_5^2, \zeta_5^3, \zeta_5^4$, if $\zeta_5 + \zeta_5^{-1} \in \mathbb{Q}$ this violates the linear independence of the basis). Further, since $\mathbb{Q}(\alpha) \subseteq \mathbb{R}$

($z \in \mathbb{C} \implies z + \bar{z} \in \mathbb{R}$ and $\zeta_5^{-1} = \bar{\zeta}$), $\mathbb{Q}(\alpha) \neq \mathbb{Q}(\zeta)$. Hence $\mathbb{Q}(\alpha)$ is our quadratic subfield. In this case, we can do better using

$$\zeta_5^4 + \zeta_5^3 + \zeta_5^2 + \zeta_5 + 1 = 0 \qquad (m_{\zeta_5,\mathbb{Q}}(x) = x^4 + x^3 + x^2 + x + 1)$$

Notice

$$\begin{aligned}\alpha^2 &= (\zeta_5 + \zeta_5^{-1})^2 \\ &= \zeta_5^2 + 2 + \zeta_5^{-1} \\ &= \zeta_5^2 + 2 + \zeta_5^3\end{aligned}$$

so

$$\alpha^2 + \alpha = \zeta_5^3 + 2 + \zeta_5^3 + \zeta_5 + \zeta_5^4$$

Hence, $\alpha^2 + \alpha - 1 = 0$, so

$$\alpha = \frac{-1 \pm \sqrt{1+4}}{2} = \frac{-1 \pm \sqrt{5}}{2} = -\frac{1}{2} \pm \frac{\sqrt{5}}{2} \in \mathbb{Q}(\sqrt{5})$$
$$\implies \mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\sqrt{5}).$$

Finally,

$$\begin{array}{c}\mathbb{Q}(\zeta_5) \\ | \\ \mathbb{Q}(\sqrt{5}) \\ | \\ \mathbb{Q}\end{array}$$

Example 5.3.2. $\mathbb{Q}(\zeta_p)$ for an odd prime p.
We can construct a generating element for the subfields of $\mathbb{Q}(\zeta_p)$. A basis for $\mathbb{Q}(\zeta_p)$ is given by

$$1, \zeta_p, \zeta_p^2, ..., \zeta_p^{p-2}$$

since $\zeta_p^{p-1} + \zeta_p^{p-2} + ... + \zeta_p + 1 = 0$. We can deduce that $\zeta_p, \zeta_p^2, \zeta_p^3, ..., \zeta_p^{p-1}$ form a basis (or else uniqueness of $m_{\zeta_p,\mathbb{Q}}(x)$ would be violated). We use this since any $\sigma \in \mathrm{Gal}\,(\mathbb{Q}(\zeta_p)/\mathbb{Q}) = G$ will permute this basis. It is important that p is prime here since the primitive 4th-roots of unity are not linearly independent ($i, -i$ are the primitive 4th-roots of unity $e^{\frac{2\pi ik}{4}}$, $\gcd(k, 4) = 1$). Let H be a subgroup of G. Let

$$\alpha_H = \sum_{\sigma \in H} \sigma(\zeta_p)$$

For any fixed $\tau \in H$, we have

$$\tau(\alpha) = \tau\left(\sum_{\sigma \in H} \sigma(\zeta_p)\right) = \sum_{\sigma \in H} \tau\sigma(\zeta_p)$$

Since H acts on itself via left multiplication, we know that τ permutes the elements of H, so

$$\sum_{\sigma \in H} \tau\sigma(\zeta_p) = \sum_{\sigma \in H} \sigma(\zeta_p)$$

α lies in the fixed field of H. Suppose $\tau \notin H$, then

$$\tau(\alpha) = \sum_{\sigma \in H} \tau\sigma(\zeta_p)$$

Since $1 \in H$, a term in this sum is

$$\tau(1(\zeta_p)) = \tau(\zeta_p)$$

If we assume that $\tau(\alpha) = \alpha$, then since the elements in this sum are a basis we must have $\tau(\zeta_p) = \sigma(\zeta_p)$ for some $\sigma \in H$. But then

$$\sigma^{-1}\tau(\zeta_p) = \zeta_p.$$

Since automorphisms of $\mathbb{Q}(\zeta_p)$ are determined by their actions on ζ_p, we have that

$$\sigma^{-1}\tau = 1 \implies \tau = \sigma \in H,$$

which is a contradiction. So α is not fixed by any automorphisms not contained in H. So, $\mathbb{Q}(\alpha)$ is precisely the fixed field of H.

Example 5.3.3. Subfields of $\mathbb{Q}(\zeta_{13})$.
13 is prime, so

$$G = \mathrm{Gal}\left(\mathbb{Q}(\zeta_{13})/\mathbb{Q}\right) \cong \mathbb{Z}/12\mathbb{Z}$$

since $(\mathbb{Z}/13\mathbb{Z})^\times$ is cyclic. 2 generates $(Z/13Z)^\times$, hence σ_2 generates G and $\sigma_2(\zeta_{13}) = \zeta_{13}^2$. We have subgroups of orders

$$12, 6, 4, 3, 2, 1$$

which correspond to the generators

$$\sigma_2, \sigma_2^2, \sigma_2^3, \sigma_2^4, \sigma_2^6, \sigma_2^{12}$$

respectively. The corresponding fixed fields will have degrees

$$1, 2, 3, 4, 6, 12 \qquad \text{(index of the subgroups of } G\text{)}$$

Let's compute the α's:

$$\begin{aligned}
\sum_{\sigma_2 \in \langle \sigma^6 \rangle} &= 1(\zeta_{13}) + \sigma_2^6(\zeta_{13}) \\
&= \zeta_{13} + (\sigma_2(\zeta_{13}))^6 \\
&= \zeta_{13} + \zeta_{13}^{2^6} \\
&= \zeta_{13} + \zeta_{13}^{64} \\
&= \zeta_{13} + \zeta_{13}^{-1}
\end{aligned}$$

so the fixed field of $\langle \sigma_2^6 \rangle$ is $\mathbb{Q}(\zeta_{13} + \zeta_{13}^{-1})$. Now,

$$\begin{aligned}
\sum_{\sigma \in \langle \sigma_2^4 \rangle} \sigma(\zeta_{13}) &= 1(\zeta_{13}) + \sigma_2^4(\zeta_{13}) + \sigma_2^8(\zeta_{13}) \\
&= \zeta_{13} + \zeta_{13}^{2^4} + \zeta_{13}^{2^8} \\
&= \zeta_{13} + \zeta_{13}^3 + \zeta_{13}^9
\end{aligned}$$

The fixed field of $\langle \sigma_4 \rangle$ is $\mathbb{Q}(\zeta_{13} + \zeta_{13}^3 + \zeta_{13}^9)$.